

The resultant Westergaard stress functions after the cracks are introduced, **Fig. 12(a) and (b)**, are determined simply by superimposing **Eq. (71)** on **Eqs. (86a) and (86b)**.

$$(a) \quad Z_I(z) = \frac{\sigma}{\sqrt{1 - (a/z)^2}}, \quad Z_{II}(z) = 0 \quad (88a)$$

$$(b) \quad Z_I(z) = \frac{\sigma}{\sqrt{1 - (a/z)^2}}, \quad Z_{II}(z) = -i\frac{\sigma}{2} \quad (88b)$$

However, adjustment for σ_x based on Z_I alone is often made in the manner described in the paragraph following **Eq. (71)** without introducing Z_{II} .

An additional example of nonzero Z_{II} in Mode I field and an example of nonzero Z_I in Mode II field are given next.

(c) An infinite plate subjected to in-plane bending $\sigma_y = cx$ at infinity (**Fig. 13**):

$$Z_I(z) = cz, \quad Z_{II}(z) = -i\frac{c}{2}z; \quad \Phi = \frac{1}{6}cx^3 \quad (89)$$

(d) An infinite plate subjected to uniform shear $\tau_{xy} = \tau$ at infinity (**Fig. 14**):

$$Z_{II}(z) = \tau, \quad Z_I(z) = i\tau; \quad \Phi = -\tau xy \quad (90)$$

Note that $Z_{II} = \tau$ alone yields the correct stress field and therefore the correct Airy stress function Φ . The presence of nonzero Z_I may seem trivial. However, although Z_I does not contribute to Φ [$\Phi_I = \text{Re}\bar{Z}_I + y \text{Im}\bar{Z}_I$ in **Eq. (83)** cancels out], Z_I does contribute to displacements, **Eqs. (85)**, and Z_{II} and Z_I of **Eq. (90)** together yield the correct elastic field (symmetric with respect to $y = \pm x$).

Again, Z_{II} in (c) and Z_I in (d) have no significance in the analysis of cracks.

C. Effect of Surface Interference of Partly Closed Cracks

In Mode I displacement field, when there is a crack-tip with a negative K_I , the crack opening displacement near that tip is also negative and thus the material would "overlap." Such overlapping is physically unacceptable and, consequently, solutions involving negative K_I are not valid by themselves. However, these negative K_I and negative openings can be directly superimposed on the positive values resulting from other applied loads, as long as the resultant K_I remains positive and the surface overlapping is eliminated. In the subsequent solution pages, the effect of crack surface interference is ignored for the most part, and negative K_I and negative opening displacements are given as solutions. These negative values, therefore, must be treated accordingly.

When the surfaces of cracks containing the tips with negative K_I entirely close, the resulting geometric and loading configuration would be obvious, and the analysis can be made in a usual manner for the final crack geometry.

Examples of such situations are:

a. A double-edge cracked strip under in-plane bending (**page 11.5**)

The resulting configuration is effectively a single-edge cracked strip with a crack on the tension side only, **Fig. 15a (page 2.13)**.

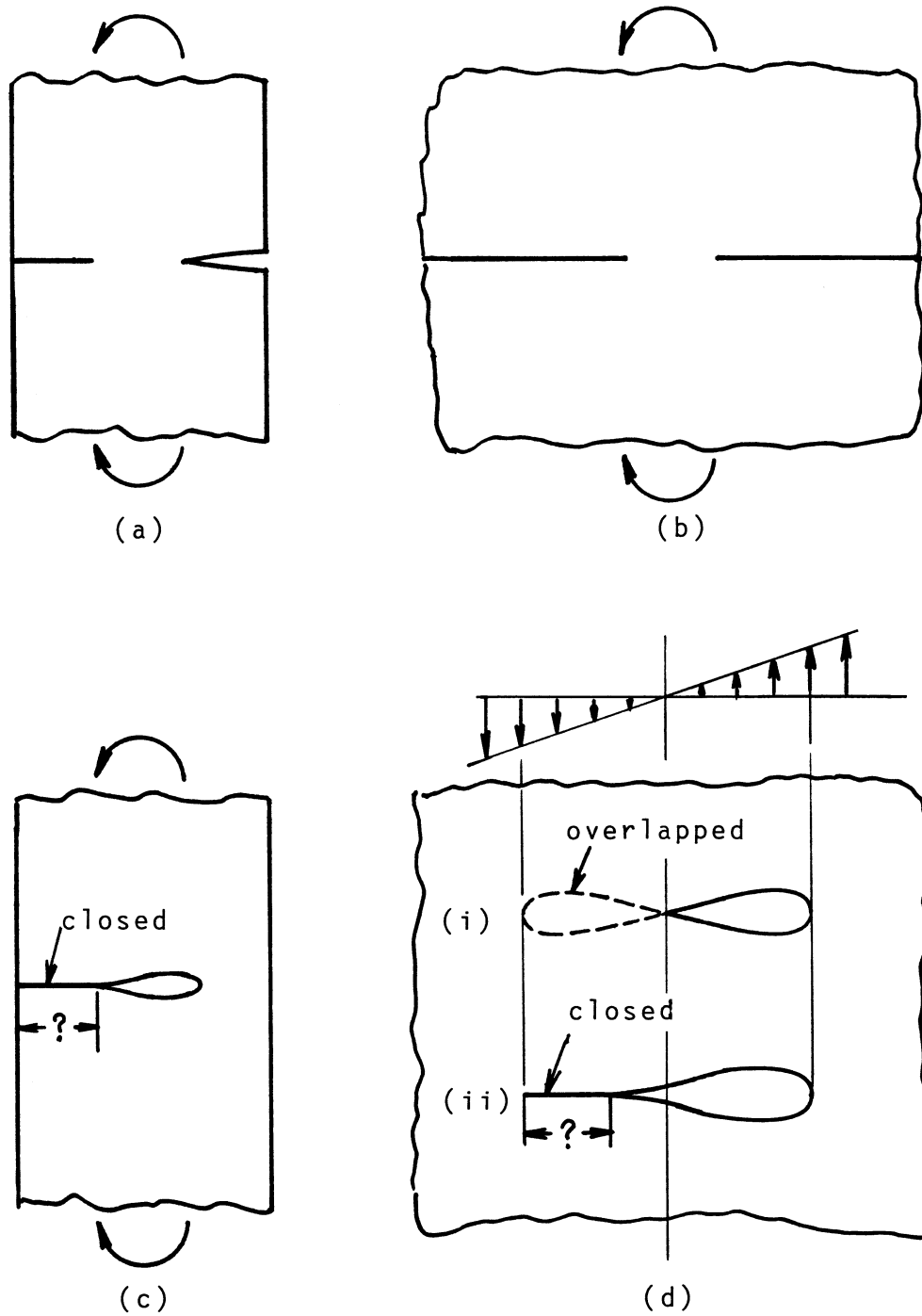


Fig. 15

- b. An infinite plate with opposing semi-infinite cracks (leaving a finite net ligament) subjected to a finite in-plane bending moment (**page 4.10**)

When the crack on the compression side totally closes, the crack on the tension side also loses its significance. The resulting configuration is simply an unloaded plate, **Fig. 15b**.

For these cases, no further discussion is needed here.

When, on the other hand, the crack surfaces only partially close, the interference of closure does affect the resulting geometric configuration and thus the overall elastic field. The geometry of such a crack is not readily knowable beforehand. For example, the length of the closed portion of the crack cannot be determined without analysis.

Examples of such situations are:

- c. A strip with a very deep single-edge crack under in-plane bending (**page 2.13** with the bending moment reversed) because the crack-tip is now on the tension side, **Fig. 15(c)**, the crack surfaces on the compression side close smoothly ($K_I = 0$) only partly, and the edge-cracked strip becomes effectively a strip with an internal crack, the resulting geometry of which is not immediately known, **Fig. 15(c)**.
- d. An infinite plate with a finite crack subjected to in-plane bending about the symmetric axis of the crack (i.e., linearly varying σ_y at infinity) (**pages 5.18/18a/18b**). (i) in **Fig. 15(d)** is the crack profile of **page 5.18a** where the effect of the crack surface contact is ignored. That is, K_I at the left tip is negative and the left half of the crack surfaces overlap. As discussed earlier, this solution may be directly superimposed on solutions for other loadings when the resulting configurations are physically acceptable ($K_I \geq 0$ at the left tip and no surface overlapping). (ii) is the crack profile when the effect of crack surface contact is accounted for (**page 5.18b**, which is to be obtained subsequently). Again, the length of the closed portion and therefore the final geometry of the crack is not known beforehand. Consequently, before determining (by superposition in particular) K_I at the right tip and the crack opening profile, and so on, including the effect of surface interference, the final crack geometry must be known.

For some analyses and discussions on the effect of surface interference of partly closed cracks, see **Seeger (1973)**, **Paris (1975b)**, **Bowie (1976a,b)**, and **Gustafson (1976)**. For additional discussions related to such surface contact, see **Westergaard (1939)** and also **Appendix G** of this handbook.

Next, the effect of crack surface interference is analyzed in detail for example (d). From the analysis of this simple example, various general characteristics of the effect of surface contact, and some features specific to the example, are observed, some of which are well known, or physically or intuitively obvious. The approach is in essence to focus on the portion of the crack that remains open and to remove the stress singularity (K_I) at the point of separation of the closed surfaces by using the usual superposition method.

Let us take, for convenience, the (x, y) coordinate system and $z = x + iy$ with the origin at the midpoint of the crack which remains open, as shown in **Fig. 16**. Other quantities specifying the geometric and loading configurations are also defined in **Fig. 16**. The coordinate system and dimensions associated with the original configuration of the crack are indicated with a subscript "0." The right crack-tips in the two systems, $x = a$ and $x_0 = a_0$, are common, but the position of the left tip, $x = -a$, under loading is not known.

The position of the left tip, that is, the point of separation of surfaces in contact, $x = -a$, is determined from the condition that the resultant K_I vanishes at this point. Referring to **Fig. 16**, the linearly varying stress $\sigma_y = p(x_0/a_0)$ in the original system, line (A), is obtained by superposition of a linearly varying stress $\sigma_y = p'(x/a)$, line (B), and a uniform stress $\sigma_y = p''(p' + p'' = p)$. Therefore, all that is required for the analysis are the solutions found on **pages 5.1/1a and 5.18/18a** applied to the opened portion, $-a \leq x \leq a$.

The resultant K_I are, by superposition,

$$K_{I=\pm a} = \pm \left(p' / 2 \right) \sqrt{\pi a} + p'' \sqrt{\pi a} \quad (91)$$

$K_{Ix=-a} = 0$ gives

$$p'' = p'/2 \quad (92a)$$

and correspondingly

$$p' = 2p/3 \quad (92b)$$

$$K_{Ix=+a} = p' \sqrt{\pi a} = (2p/3) \sqrt{\pi a} \quad (92c)$$

In addition, the length of closure

$$\ell = a = 2a_0/3 \quad (92d)$$

is now obvious from the geometry of **Fig. 16**. That is, the left one-third of the original crack closes or, in other words, the one-third of the crack on the compression side remains open ($a_c = a_t = a_0/3$). These relations are summarized at the bottom of **Fig. 16**.

The complete set of expressions resulting from superposition of **pages 5.1/1a and 5.18/18a** after incorporating the preceding results are now readily obtained. They are summarized in terms of $z = x + iy$ and a in **Fig. 16** as follows:

$$Z_I(z) = \frac{p}{3a} (2z - a) \sqrt{\frac{z+a}{z-a}} \quad (93a)$$

$$\bar{Z}_I(z) = \frac{p}{3a} (z+a) \sqrt{(z+a)(z-a)} \quad (93b)$$

$$K_{Ix=-a} = 0, \quad K_{Ix=+a} = \frac{2}{3} p \sqrt{\pi a} \quad (93c)$$

$$\sigma_y(x, 0) = \frac{p}{3a} (2x - a) \sqrt{\frac{x+a}{x-a}} \quad (93d)$$

$$2v(x, 0) = \frac{4pa}{3E'} \left(1 + \frac{x}{a}\right) \sqrt{1 - \left(\frac{x}{a}\right)^2} \quad (93e)$$

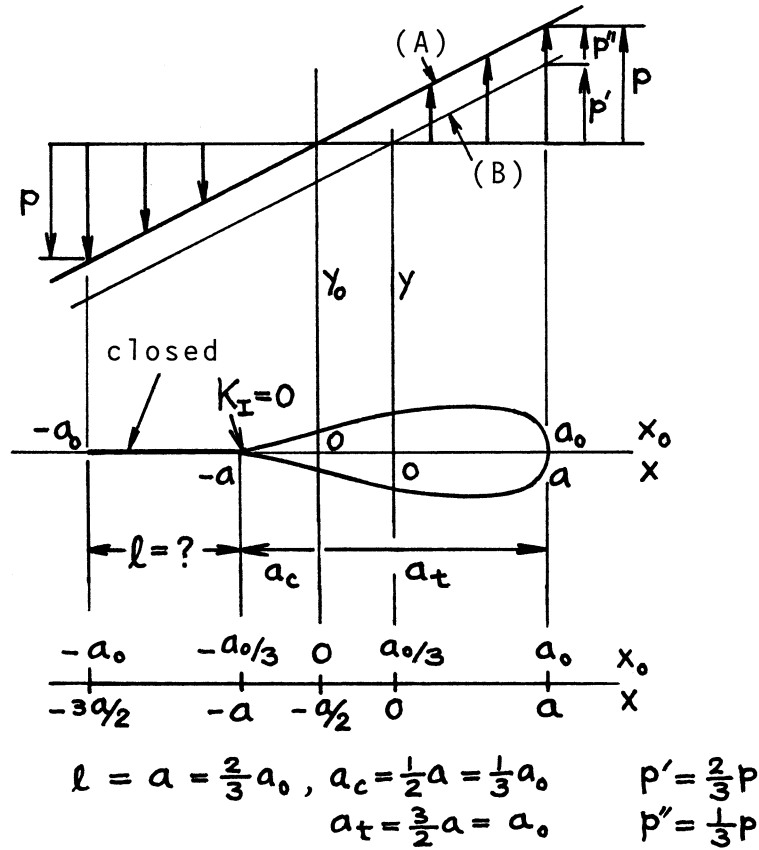


Fig. 16

$$\text{Area of Crack: } A = \frac{2\pi}{3} \frac{pa^2}{E'} \quad (93f)$$

These expressions are converted to those in terms of the original system $z_0 = x_0 + iy_0$ and a_0 by replacing a by $2a_0/3$, **Eq. (92d)**, and z by $z_0 - a/2 = z_0 - a_0/3$, then omitting the subscript "0." The complete expressions corresponding to **Eqs. (93)** and the resulting geometry of the crack are presented on **page 5.18b**. **Page 5.18b** is directly compared with **page 5.18a** for the effect of surface interference. Discussions of the comparison will follow.

The following observations are based on the results of the analysis of the present example. Some of them are general characteristics of surface interference and others are specific features of the present example.

- The stress variation on the closed portion of the crack near the point of separation of the closed surfaces is in the form of a parabola ($\propto r^{1/2}$), and the separation profile of the surfaces near that point is in the form of a semicubical parabola ($\propto r^{3/2}$), as is well known (**Westergaard 1939**).
- Once crack surfaces close and remain closed, the presence or absence of cracks beyond the point of separation is immaterial in the subsequent analysis of cracks that remain open.

- c. $K_I = 0$ at a tip ensures finite stresses and a smooth separation of the surfaces at that tip. However, $K_I = 0$ by itself does not necessarily require the stresses to be zero at the tip, as observed from the preceding analysis (e.g., **page 5.18b**, " $\sigma_y(x, 0)$ - effect of crack" indicated by a dashed line) and the (Dugdale) yield strip model analysis (**pages 30.1 - 32.6**). See f. below for further discussion.
- d. The crack surface interference naturally increases K_I at the open end of the crack and the crack opening area. For the present example, K_I is increased to $2(2/3)^{3/2} = 1.089$ times the corresponding value with the surface overlapping permitted, and the crack opening area becomes $4\pi/9 = 1.396$ times that of the opened (right) half. See **pages 5.18a and 5.18b**.
- e. A crack-tip located in a compressive region does not necessarily close; that is, K_I remains positive if the other tip remains open $K_I > 0$. For the present example of linearly varying σ_y in **Fig. 16**, when the left tip is located in $x_0 > -a_0/3$, obviously $K_I > 0$, and therefore the left tip located in the compression region $-a_0/3 < x_0 < 0$ will remain open.
- f. As discussed in b., the overall solution in terms of $Z(z)$ after cracks are introduced is given (for Mode I) by

$$\left. \begin{aligned} Z_I(z) &= Z_I(z)_{\text{no crack}} + Z_I(z)_{\text{due to crack}} \\ Z_{II}(z) &= Z_{II}(z)_{\text{no crack}} \end{aligned} \right\} \quad (94)$$

where $Z_I(z)_{\text{due to crack}}$ is determined by the integral of **Eq. (70)** or the corresponding integral for other crack configurations. As observed from the example of **Eq. (71)**, the integrals generally result in

$$\left. \begin{aligned} Z_I(z)_{\text{due to crack}} &= Z_I(z)_{\text{function of geometry}} - Z_I(z)_{\text{no crack}} \\ Z_{II}(z)_{\text{due to crack}} &= 0 \end{aligned} \right\} \quad (95)$$

Thus, the resultant field, **Eq. (94)**, is always

$$\left. \begin{aligned} Z_I(z) &= Z_I(z)_{\text{function of crack geometry}} \\ Z_{II}(z) &= Z_{II}(z)_{\text{no crack}} \end{aligned} \right\} \quad (96)$$

$Z_{II}(z)$ may be disregarded here because it has nothing to do with the presence of the cracks.

Therefore, it should be noted that the solution given on **page 5.18** is strictly for the overall situation of **page 5.18a**. That is, for **page 5.18**, $Z_I(z)_{\text{no crack}} (= pz/a)$ and $Z_{II}(z)_{\text{no crack}} (= -ipz/2a)$ [**Eq. (89)** with $c = p/a$] are disregarded.

Similarly, for the present example of **page 5.18b**, the integral of **Eq. (70)** results in

$$\left. \begin{aligned} Z_I(z)_{\text{due to crack}} &= \frac{p}{a} \left(z - \frac{2a}{3} \right) \sqrt{\frac{z+a/3}{z-a}} - \frac{p}{a} z \\ Z_{II}(z)_{\text{due to crack}} &= 0 \end{aligned} \right\} \quad (97)$$

and correspondingly

$$\sigma_y(-a/3, 0) = p/3 \neq 0 \quad (98)$$

although K_I at the cuspidal tip, $x = -a/3$, is zero, as discussed in c. above. In addition, it may be interesting to note that

$$\begin{aligned} \sigma_y(x, 0) &> 0 \\ x &\leq -a/3 \end{aligned} \quad (99)$$

that is, $\sigma_y(x, 0)$ on the closed portion of the crack, $-a \leq x \leq -a/3$, and beyond the left tip, $x < -a$, is in tension rather than compression until the remote stress is superimposed. Refer to the distribution of $\sigma_y(x, 0)$, $x \leq -a/3$, found on **page 5.18b**.

The final result presented on **page 5.18b** is the superposition of **Eq. (97)** and **Eq. (89)** with $c = p/a$, that is,

$$Z_I(z) = \frac{p}{a} \left(z - \frac{2a}{3} \right) \sqrt{\frac{z + a/3}{z - a}}, \quad Z_{II}(z) = -i \frac{p}{2a} z \quad (100)$$

$Z_{III}(z)$ is not included on **page 5.18b**.

- g. It is obvious that cracks having tips with negative K_I totally close, partially close with cuspidal ends, or remain open. It is also obvious, from the analysis of the example, that the final geometry of cracks is unique; that is the positions of the cuspidal ends are uniquely determined as long as the applied load is proportional, regardless of the level of the applied load. The "proportional loading" here is in a sense similar to that used in the theory of plasticity, but less restrictive. It is sufficient if $\sigma_y(x, 0)$ in the absence of cracks is proportional (actually, if it is proportional only on the segments corresponding to the portions of cracks that eventually remain open). For the linearly varying σ_y at infinity (in-plane bending), the proportional loading is realized when the position of zero crossing of σ_y (i.e., the axis of moment) is fixed. Referring to **Fig. 17**, let us assume for simplicity that only one right crack-tip is on the tension side. **Fig. 17a** is obvious; both tips of the right-most crack are in the tensile region. The presence of cracks in the compressive region is immaterial. In **Fig. 17b**, the left tip of the crack is in the compressive region, but the length on the compression side is less than one-third of that on the tension side ($a_c < a_t/3$), and both tips still remain open, as discussed in e. above. The presence of cracks ahead of the left tip is immaterial. When $a_c \geq a_t/3$, **Fig. 17c**, a cuspidal end is always formed at $x = -a_t/3$. That is, its position is uniquely determined by the length on the tension side, regardless of the level of the applied load, or the presence or absence of cracks in $x \leq -a_t/3$. Only the crack opening profile changes proportionally to the applied load.
- h. The analysis of the crack surface interference in three-dimensional configurations will be much more involved and would require a numerical approach.

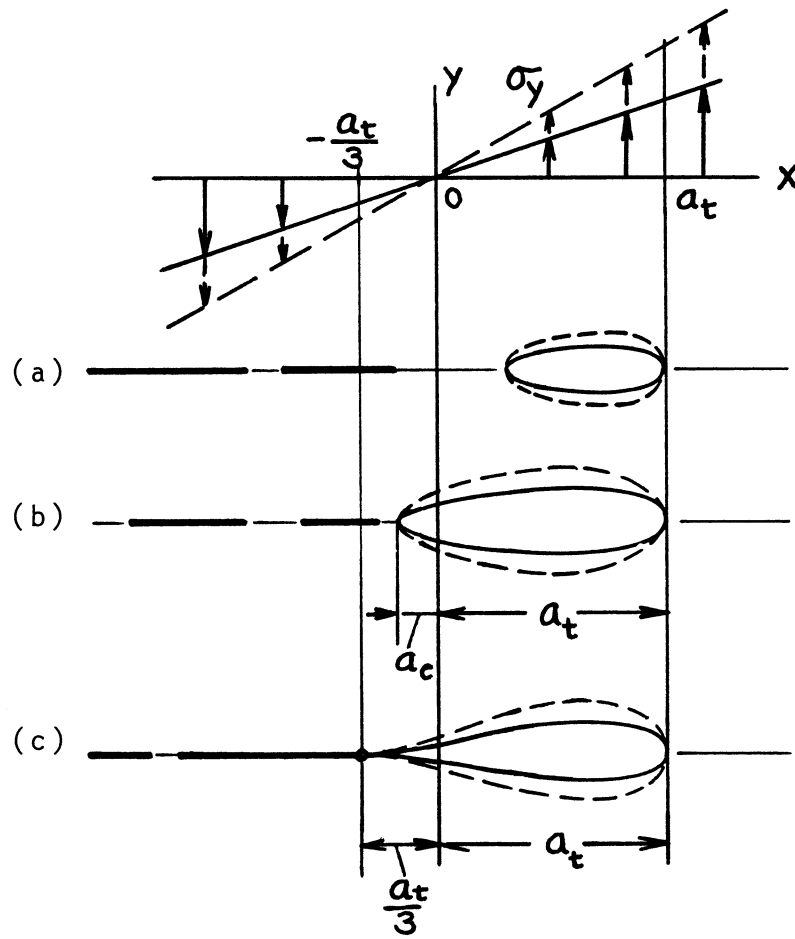


Fig. 17

STRESS ANALYSIS RESULTS FOR COMMON TEST SPECIMEN CONFIGURATIONS

- ☐ The Center Cracked Test Specimen
- ☐ The Double Edge Notch Test Specimen
- ☐ The Single Edge Notch Test Specimen
- ☐ The Pure Bending Specimen
- ☐ The Three-Point Bend Test Specimen
- ☐ The Compact Tension Test Specimen
- ☐ The Round (Disc-Shaped) Compact Specimen
- ☐ The Arc-Shaped (C-Shaped) Specimen
- ☐ Other Common Specimen Configurations
- ☐ Electrical Potencial Calibraton

THE CENTER CRACKED TEST SPECIMEN

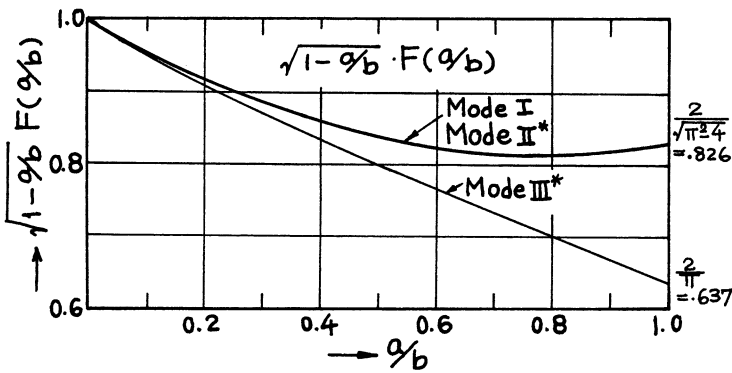
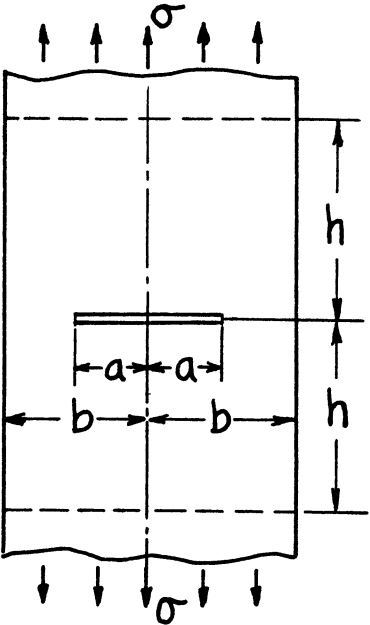
A. Stress Intensity Factor

$$K_I = \sigma \sqrt{\pi a} F(a/b)$$

Numerical Values of $F(a/b)$

(Isida 1962, 1965a, b, 1973)

Isida's 36-term power series of $(a/b)^2$ (Laurent series expansion of complex stress potential, 1973) gives practically exact values of $F(a/b)$ up to $a/b = 0.9$. Numerical values of $F(a/b)$ are shown in the following graph and table.



a/b	$F(a/b)$
0.0	1.0000
0.1	1.0060
0.2	1.0246
0.3	1.0577
0.4	1.1094
0.5	1.1867
0.6	1.3033
0.7	1.4882
0.8	1.8160
0.9	2.5776
1.0	$\frac{2}{\sqrt{\pi^2-4}} \sqrt{1-a/b}^{**}$

**Exact Limit (Koiter 1965b)

*See Note 2

Empirical Formulas

- a. Accuracy
- b. Method of derivation, reference

$$F(a/b) = \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}}$$

- a. Better than 5% for $a/b \leq 0.5$
- b. Approximation by periodic crack solution (**Irwin 1957**)

$$F(a/b) = 1 + 0.128(a/b) - 0.288(a/b)^2 + 1.525(a/b)^3$$

- a. 0.5% for $a/b \leq 0.7$
- b. Least squares fitting to Isida's results (**Brown 1966**)

$$F(a/b) = \sqrt{\sec \frac{\pi a}{2b}}$$

- a. 0.3% for $a/b \leq 0.7$, 1% at $a/b = 0.8$
- b. Guess based on Isida's results (**Feddersen 1966**)

$$F(a/b) = \frac{1 - 0.5(a/b) + 0.326(a/b)^2}{\sqrt{1 - a/b}}$$

- a. 1% for any a/b
- b. Asymptotic approximation (**Koiter 1965b**)

$$F(a/b) = \frac{1 - 0.5(a/b) + 0.370(a/b)^2 - 0.044(a/b)^3}{\sqrt{1 - a/b}}$$

- a. 0.3% for any a/b
- b. Modification of Koiter's formula (**Tada 1973**)

$$F(a/b) = \left\{ 1 - 0.025(a/b)^2 + 0.06(a/b)^4 \right\} \sqrt{\sec \frac{\pi a}{2b}}$$

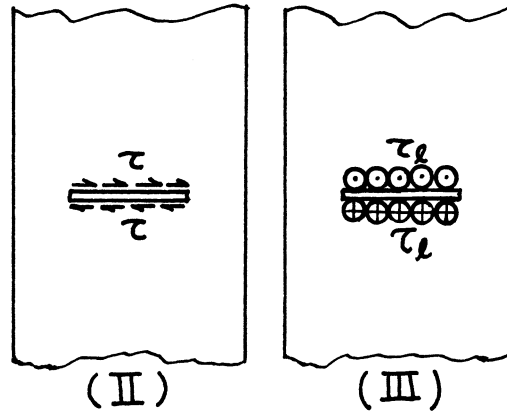
- a. 0.1% for any a/b
- b. Modification of Feddersen's formula (**Tada 1973**)

- NOTES: 1. Finite height configuration (h/b finite) is given separately. When $h/b \geq 3$, the plate is practically regarded as an infinite strip as far as the effects of h/b on K are concerned (**Isida 1971a**).
2. For Mode II configuration (II), the correction factor is identical to $F(a/b)$ in Mode I.

$$K_{II} = \tau \sqrt{\pi a} F(a/b)$$

For Mode III configuration (III), the following formula is exact:

$$K_{III} = \tau_\ell \sqrt{\pi a} \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}}$$



Other Methods and References

1. Compliance Method: **Forman 1964**
2. Fourier Transform - Integral Equation: **Sneddon 1971b**
3. Finite Element Method: **Mendelson 1972, Yamamoto 1972**
4. Boundary Collocation Method: **Bowie 1970a**
5. Integral Equations - Successive Stress Relaxation: **Tada 1971, 1972a, b**

(See also pages 2.24, 2.26, 2.35, 2.36, 7.1, 11.1, 11.2, 11.3, 11.4, 18.1, 18.2, 18.3, 19.4, 20.1, 20.2, 20.3, etc., for related solutions and corrections for various effects.)

B. Displacements

Crack Opening at Center

$$\delta = \frac{4\sigma a}{E'} V_1(a/b)$$

The following formula has better than 0.6% accuracy for any a/b .

$$V_1(a/b) = -0.071 - 0.535(a/b) + 0.169(a/b)^2 - 0.090(a/b)^3 \\ + 0.020(a/b)^4 - 1.071 \frac{1}{a/b} \ln(1 - a/b)$$

Additional Displacement at Remote Points ($h/b \geq 3$ Due to Presence of Crack

$$\Delta_{crack} = \Delta_{total} - \Delta_{No\ crack} = \Delta_{total} - \frac{\sigma}{E} \cdot 2h$$

$$\Delta_{crack} = \frac{4\sigma a}{E'} V_2(a/b)$$

Crack Opening Area

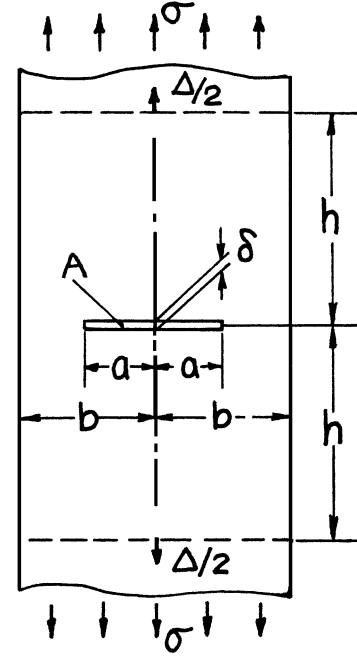
$$A = \Delta_{crack} \cdot (2b)$$

The following formula has better than 0.6% accuracy for any a/b .

$$V_2(a/b) = -1.071 + 0.250(a/b) - 0.357(a/b)^2 + 0.121(a/b)^3 - 0.047(a/b)^4 + 0.008(a/b)^5 - 1.071 \frac{1}{(a/b)} \ln(1 - a/b)$$

Method of Derivation: Paris' equation based on energy principles (Paris 1957) (see Appendix B)

Reference: Tada 1973



- NOTES: 1. $E' = E$ for plane stress and $E' = E/(1 - \nu^2)$ for plane strain.
 2. Uniform pressure σ applied on the crack surfaces results in the same crack opening δ and remote displacement Δ_{crack} .
 3. Limiting values of $V_1(a/b)$ and $V_2(a/b)$ at $a/b \rightarrow 1$ are exact.

$$V_1(a/b \rightarrow 1) = V_2(a/b \rightarrow 1) = \frac{2\pi}{\pi^2 - 4} \ln \frac{1}{1 - a/b}$$

4. For Mode II loading (II), the displacements (II') are

$$\delta = \frac{4\tau a}{E'} V_1(a/b)$$

$$\Delta = \frac{4\tau a}{E'} V_2(a/b)$$

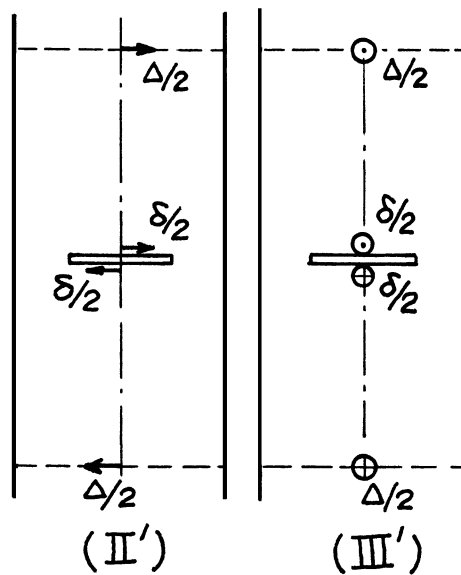
where V_1 and V_2 are identical to those in Mode I. For Mode III loading (III), the displacements (III') are given by

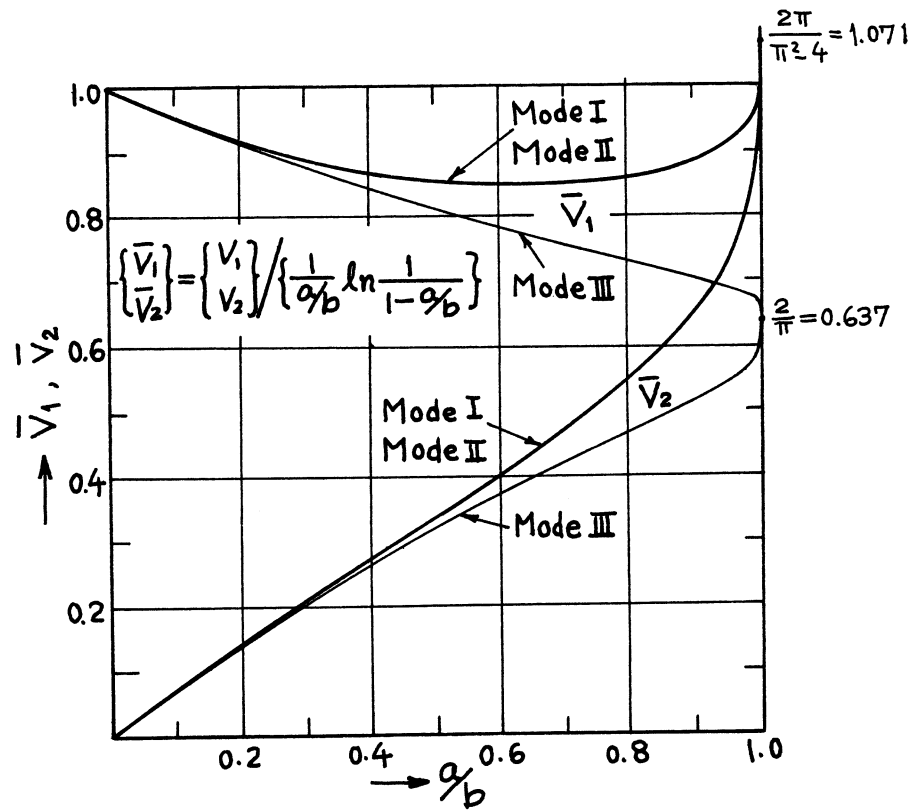
$$\delta = \frac{2\tau_t a}{G} V_1(a/b)$$

$$V_1(a/b) = \frac{2}{\pi} \frac{1}{a/b} \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

$$\Delta = \frac{2\tau_t a}{G} V_2(a/b)$$

$$V_2(a/b) = \frac{2}{\pi} \frac{1}{a/b} \ln \left(\sec \frac{\pi a}{2b} \right)$$





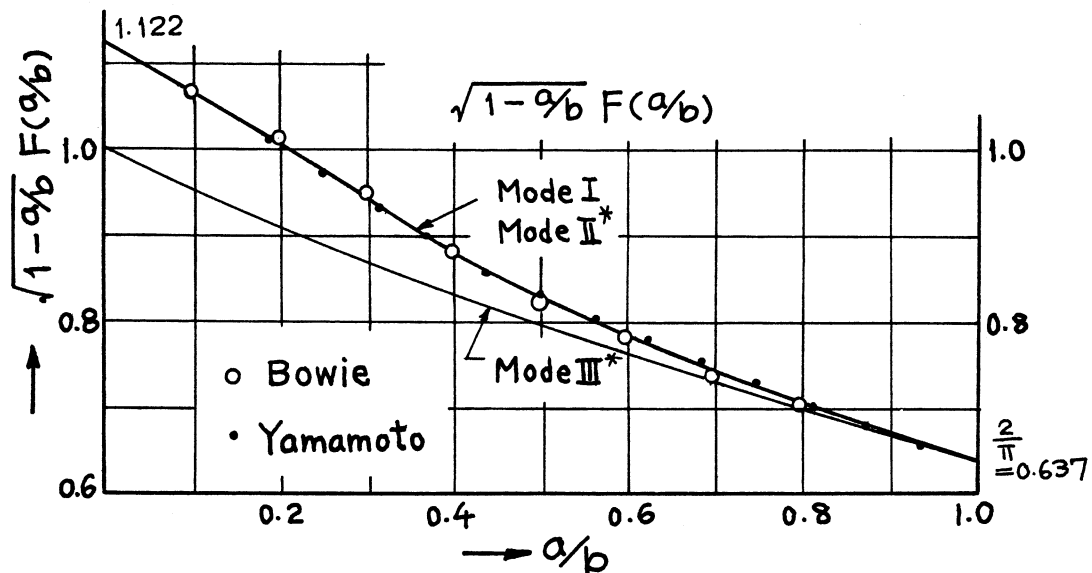
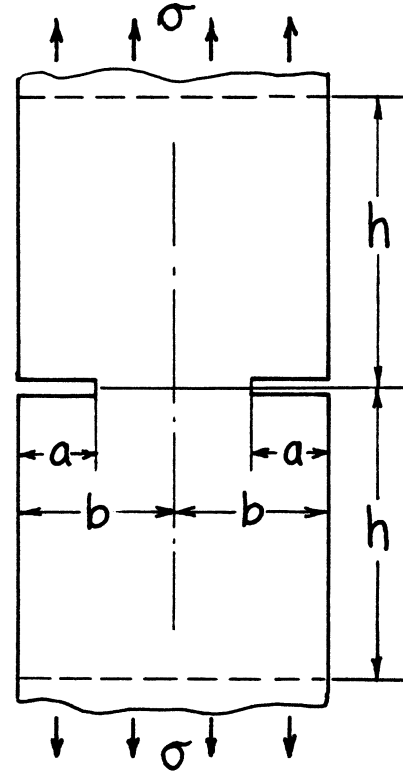
THE DOUBLE EDGE NOTCH TEST SPECIMEN

A. Stress Intensity Factor

$$K_I = \sigma \sqrt{\pi a} F(a/b)$$

Numerical Values of $F(a/b)$

Bowie's results ($h/b = 3.0$, mapping function method) have 1% accuracy and Yamamoto's results ($h/b = 2.75$, finite element method) have 0.5% accuracy for $0.2 < a/b < 0.9$ (Bowie 1964a; Yamamoto 1972).



*See Note 2

(See also pages 2.32, 2.33, 2.34, 11.5, 15.1 etc., for corrections and various effects.)

Empirical Formulas

- a. Accuracy
- b. Method of derivation, reference

$$F(a/b) = \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}}$$

- a. Better than 5% for $a/b > 0.4$
- b. Approximation by periodic crack solution (**Irwin 1957**)

$$F(a/b) = 1.12 + 0.203(a/b) - 1.197(a/b)^2 + 1.930(a/b)^3$$

- a. Better than 2% for $a/b \leq 0.7$
- b. Least squares fitting to Bowie's results (**Brown 1966**)

$$F(a/b) = \frac{1.122 - 0.561(a/b) - 0.015(a/b)^2 + 0.091(a/b)^3}{\sqrt{1 - a/b}}$$

- a. Better than 2% for any a/b
- b. Asymptotic approximation (**Benthem 1972**)

$$F(a/b) = \left(1 + 0.122 \cos^4 \frac{\pi a}{2b}\right) \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}}$$

- a. 0.5% for any a/b
- b. Modification of Irwin's interpolation formula (**Tada 1973**)

$$F(a/b) = \frac{1.122 - 0.561(a/b) - 0.205(a/b)^2 + 0.471(a/b)^3 - 0.190(a/b)^4}{\sqrt{1 - a/b}}$$

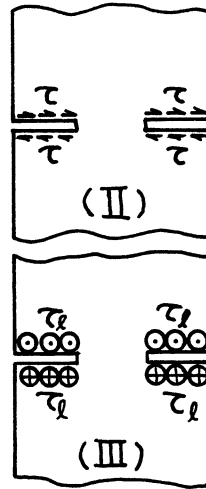
- a. 0.5% for any a/b
- b. Modification of Benthem's formula (**Tada 1973**)

- NOTE: 1. Both $h/b = 3.0$ (Bowie) and $h/b = 2.75$ (Yamamoto) are considered effectively infinite.
 2. For Mode II configuration (II), the correction factor is identical to $F(a/b)$ in Mode I.

$$K_{II} = \tau \sqrt{\pi a} F(a/b)$$

For Mode III configuration (III), the exact formula is

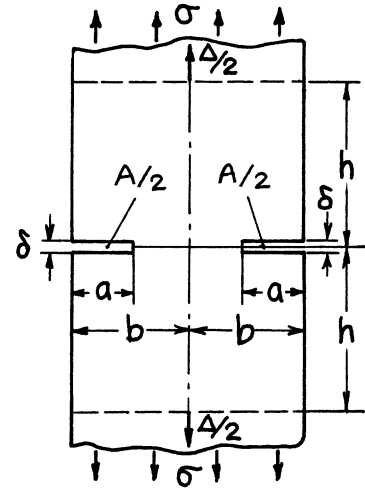
$$K_{III} = \tau_\ell \sqrt{\pi a} \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}}$$



B. Displacements

Crack Opening at Edges

$$\delta = \frac{4\sigma a}{E'} V_1(a/b)$$



The following formula has better than 2% accuracy for any a/b :

$$V_1(a/b) = \frac{1}{\left(\frac{\pi a}{2b}\right)} \left\{ 0.454 \left(\sin \frac{\pi a}{2b} \right) - 0.065 \left(\sin \frac{\pi a}{2b} \right)^3 - 0.007 \left(\sin \frac{\pi a}{2b} \right)^5 + \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right) \right\}$$

Additional Displacement at Remote Points ($h/b > 3$) Due to Presence of Cracks and Crack Opening Area

$$\Delta_{crack} = \Delta_{total} - \Delta_{no\ crack} = \Delta_{total} - \frac{\sigma}{E} \cdot 2h$$

$$\Delta_{crack} = \frac{4\sigma a}{E'} V_2(a/b)$$

$$A = \Delta_{crack} \cdot 2b$$

The following formula has better than 1% accuracy for any a/b

$$V_2(a/b) = \frac{1}{\left(\frac{\pi a}{2b}\right)} \left\{ 0.0629 - 0.0610 \left(\cos \frac{\pi a}{2b}\right)^4 - 0.0019 \left(\cos \frac{\pi a}{2b}\right)^8 + \ln \left(\sec \frac{\pi a}{2b}\right) \right\}$$

Reference: **Tada 1973**

Method of Derivation: Paris' equation based on energy principles (**Paris 1957**) (See **Appendix B**)

- NOTE: 1. $E' = E$ for plane stress
 $E' = E/(1 - \nu^2)$ for plane strain
 2. Uniform pressure σ applied on the crack surfaces results in the same crack opening δ and remote displacement Δ_{crack} .
 3. Limiting values of V_1 and V_2 at $a/b \rightarrow 0$ and $a/b \rightarrow 1$

$$V_1(a/b \rightarrow 0) = 1.454, \quad V_2(a/b \rightarrow 0) = 0$$

$$V_1(a/b \rightarrow 1) = V_2(a/b \rightarrow 1) = \frac{2}{\pi} \ell n \frac{1}{1 - a/b} \quad (\text{exact})$$

4. For Mode II loading (II), the displacements (II') are

$$\delta = \frac{4\tau a}{E'} V_1(a/b)$$

$$\Delta = \frac{4\tau a}{E'} V_2(a/b)$$

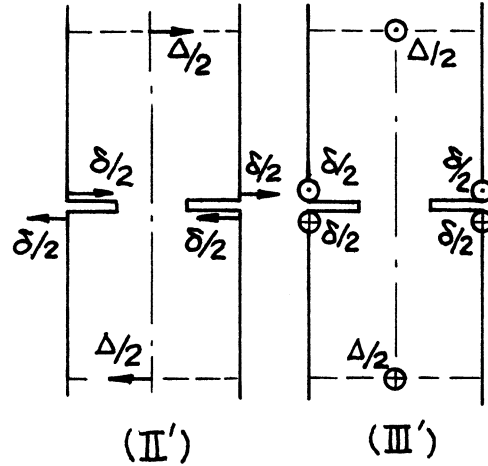
where V_1 and V_2 are identical to those in Mode I.
 For Mode III loading (III), the displacements (III') are

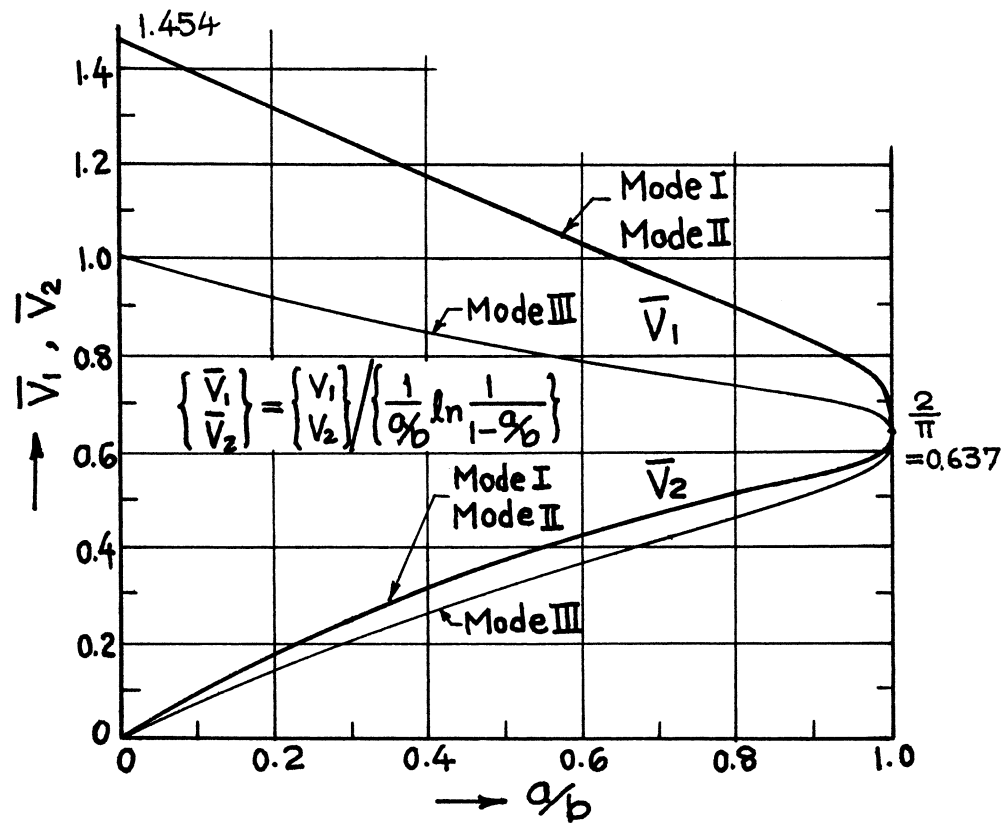
$$\delta = \frac{2\tau_\ell a}{G} V_1(a/b)$$

$$V_1(a/b) = \frac{2}{\pi} \frac{1}{a/b} \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

$$\Delta = \frac{2\tau_\ell a}{G} V_2(a/b)$$

$$V_2(a/b) = \frac{2}{\pi} \frac{1}{a/b} \ell n \left(\sec \frac{\pi a}{2b} \right)$$





THE SINGLE EDGE NOTCH TEST SPECIMEN

A. Stress Intensity Factor

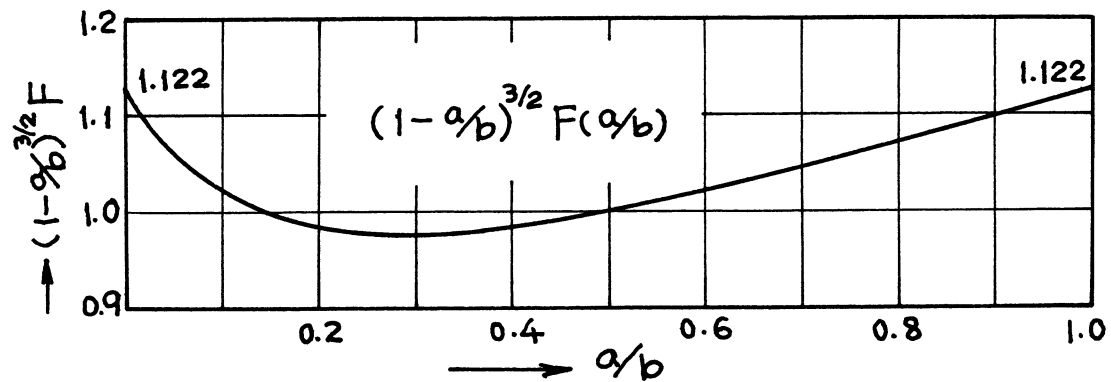
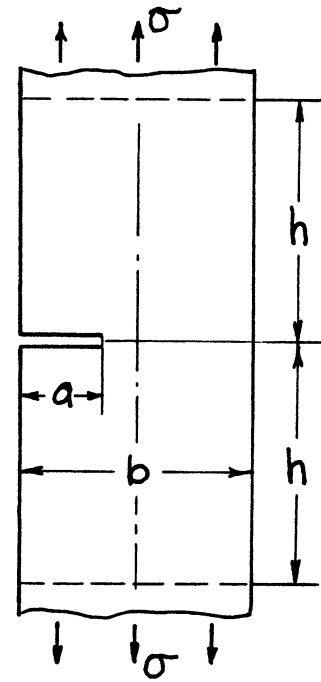
$$K_I = \sigma \sqrt{\pi a} F(a/b)$$

Numerical Values of $F(a/b)$

The curve in the following figure was drawn based on the results having better than 0.5% accuracy.

Methods and References

- Boundary Collocation Method ($h/b > 0.8$): **Gross 1964**
- Mapping Function Method ($h/b = 1.53$): **Bowie 1965**
- Green's Function Method ($h/b > 1.5$): **Emery 1969, 1972**
- Weight Function Method: **Bueckner 1970, 1971**
- Asymptotic Approximation: **Benthem 1972**
- Finite Element Method ($h/b = 2.75, 1.0$): **Yamamoto 1972**



- NOTE:
- Load is applied along the centerline of the strip at the crack location (or uniform pressure on crack surfaces).
 - The effect of h/b is practically negligible for $h/b \geq 1.0$.

(See also pages 2.13, 2.16, 2.27 to 2.31 etc., for various corrections and effects.)

Empirical Formulas

- a. Accuracy
- b. Method, reference

$$F(a/b) = 1.122 - 0.231(a/b) + 10.550(a/b)^2 - 21.710(a/b)^3 + 30.382(a/b)^4$$

- a. 0.5% for $a/b \leq 0.6$
- b. Least squares fitting (**Gross 1964; Brown 1966**)

$$F(a/b) = 0.265(1 - a/b)^4 + \frac{0.857 + 0.265 a/b}{(1 - a/b)^{3/2}}$$

- a. Better than 1% for $a/b < 0.2$, 0.5% for $a/b \geq 0.2$
- b. **Tada 1973**

$$F(a/b) = \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}} \cdot \frac{0.752 + 2.02(a/b) + 0.37 \left(1 - \sin \frac{\pi a}{2b}\right)^3}{\cos \frac{\pi a}{2b}}$$

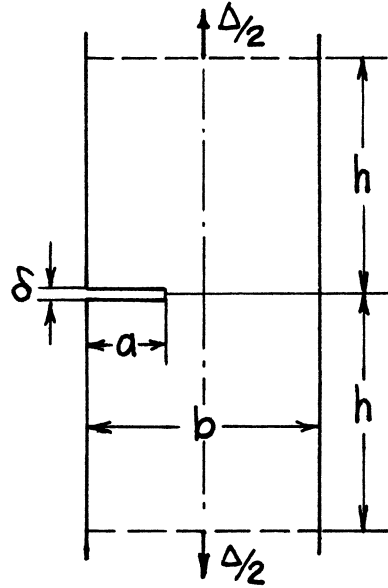
- a. Better than 0.5% for any a/b
- b. **Tada 1973**

B. Displacements**Crack Opening at Edge**

$$\delta = \frac{4\sigma a}{E'} V_1(a/b)$$

Gross' results (**Gross 1967**, Boundary Collocation Method) are expected to have 0.5% accuracy for $0.2 \leq a/b < 0.7$. An empirical formula with 1% accuracy for any a/b is (**Tada 1973**)

$$V_1(a/b) = \frac{1.46 + 3.42 \left(1 - \cos \frac{\pi a}{2b}\right)}{\left(\cos \frac{\pi a}{2b}\right)^2}$$



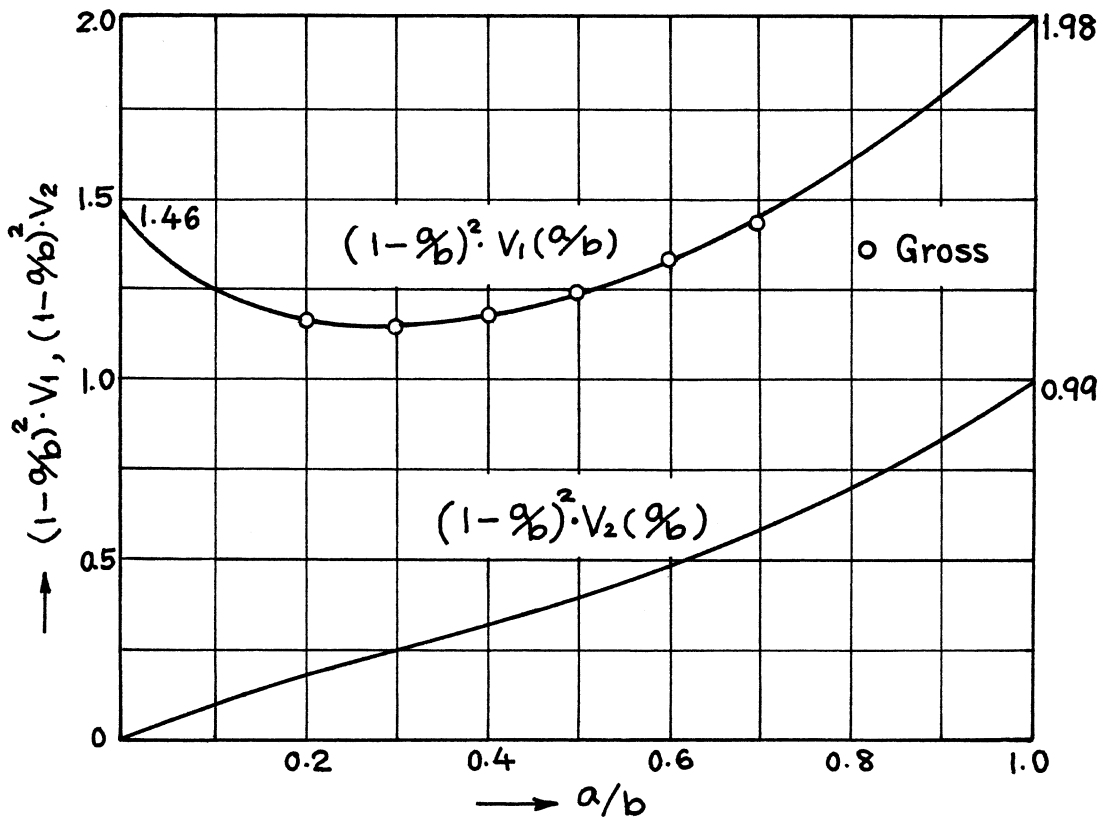
Additional Remote Point ($h/b \geq 1$) Displacement due to Crack (Along the Centerline at the Crack Location)

$$\Delta_{crack} = \Delta_{total} - \Delta_{no\ crack}$$

$$\Delta_{crack} = \frac{4\sigma a}{E'} \cdot V_2(a/b)$$

The following formula has better than 1% accuracy for any a/b :

$$V_2(a/b) = \frac{a/b}{(1-a/b)^2} \left\{ 0.99 - a/b(1-a/b) \left(1.3 - 1.2 a/b + 0.7 (a/b)^2 \right) \right\}$$



Method: Paris' Equation (Paris 1957) (See Appendix B.)

Reference: Tada 1973

THE PURE BENDING SPECIMEN

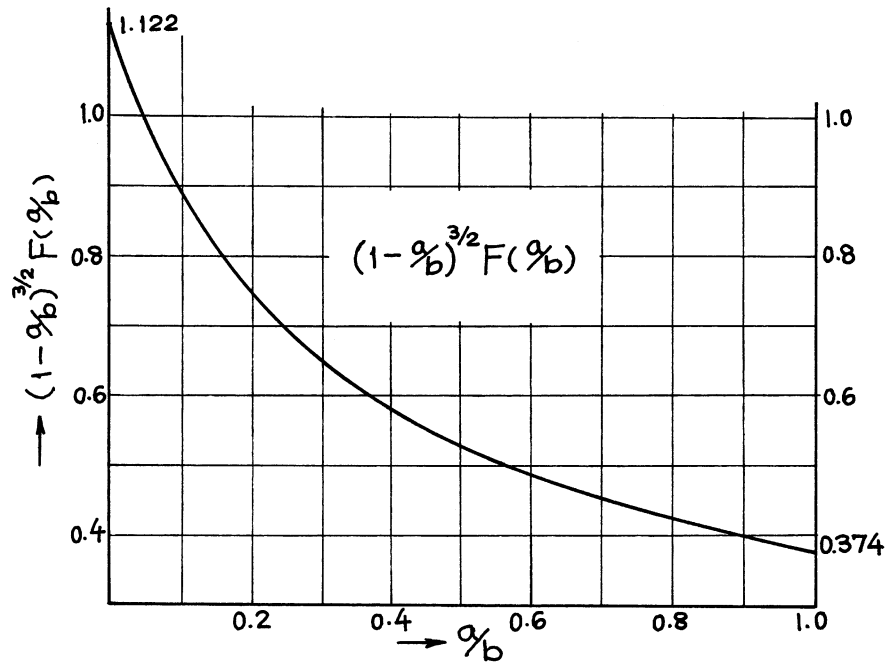
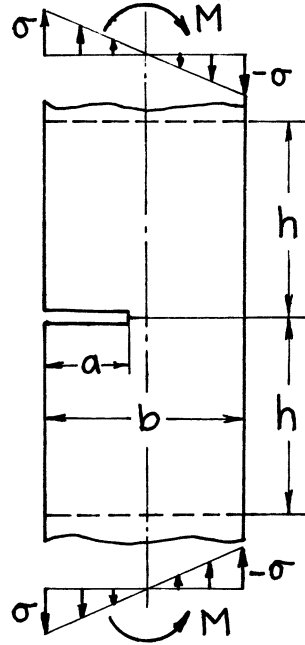
A. Stress Intensity Factor

$$\sigma = \frac{6M}{b^2}$$

$$K_I = \sigma \sqrt{\pi a} F(a/b)$$

Numerical Values of $F(a/b)$

The curve in the following figure was drawn based on the results having better than 0.5% accuracy. Also used for four-point bending.



Methods and References

1. Singular Integral Equation, **Bueckner 1960**
2. Boundary Collocation Method ($h/b \geq 2$), **Gross 1965a**
3. Weight Function Method, **Bueckner 1970, 1971**
4. Green's Function Method ($h/b \geq 1.5$), **Emery 1969**
5. Asymptotic Approximation, **Benthem 1972**

Empirical Formulas

- a. Accuracy
- b. Method, reference

$$F(a/b) = 1.122 - 1.40(a/b) + 7.33(a/b)^2 - 13.08(a/b)^3 + 14.0(a/b)^4$$

- a. 0.2% for $a/b \leq 0.6$
- b. Least squares fitting (**Brown 1966**)

$$F(a/b) = \sqrt{\frac{2b}{\pi a} \tan \frac{\pi a}{2b}} \frac{0.923 + 0.199 \left(1 - \sin \frac{\pi a}{2b}\right)^4}{\cos \frac{\pi a}{2b}}$$

- a. Better than 0.5% for any a/b
- b. **Tada 1973**

B. Displacements**Crack Opening at Edge**

$$\delta = \frac{4\sigma a}{E'} V(a/b)$$

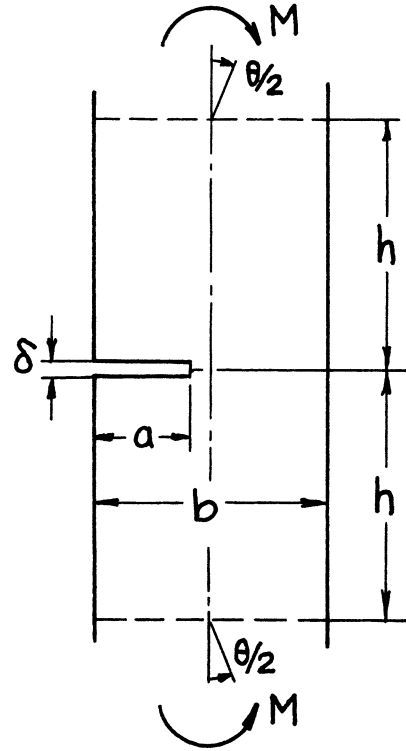
Gross' results (**Gross 1967**, Boundary Collocation Method) are expected to have 0.5% accuracy for $0.2 \leq a/b \leq 0.7$. An empirical formula with 1% accuracy for any a/b is (**Tada 1973**)

$$V(a/b) = 0.8 - 1.7(a/b) + 2.4(a/b)^2 + \frac{0.66}{(1 - a/b)^2}$$

Additional Remote Point ($h/b > 2$) Displacement (Rotation) Due to Crack

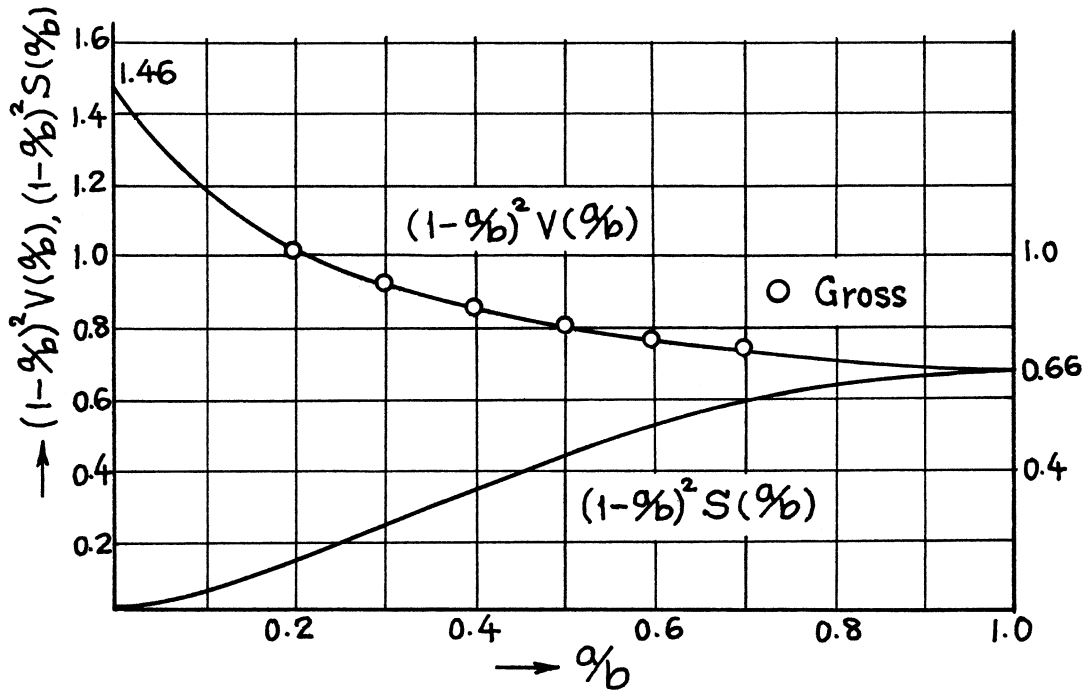
$$\theta_{crack} = \theta_{total} - \theta_{no crack}$$

$$\theta_{crack} = \frac{4\sigma}{E'} S(a/b)$$



The following formula has better than 1% accuracy for any a/b .

$$S(a/b) = \left(\frac{a/b}{1-a/b} \right)^2 \left\{ 5.93 - 19.69(a/b) + 37.14(a/b)^2 - 35.84(a/b)^3 + 13.12(a/b)^4 \right\}$$



Method: Paris' Equation (Paris 1957) (See Appendix B.)

Reference: Tada 1973

(See also pages 2.16, 2.27, 9.1 etc., for related solutions.)

THE THREE-POINT BEND TEST SPECIMEN

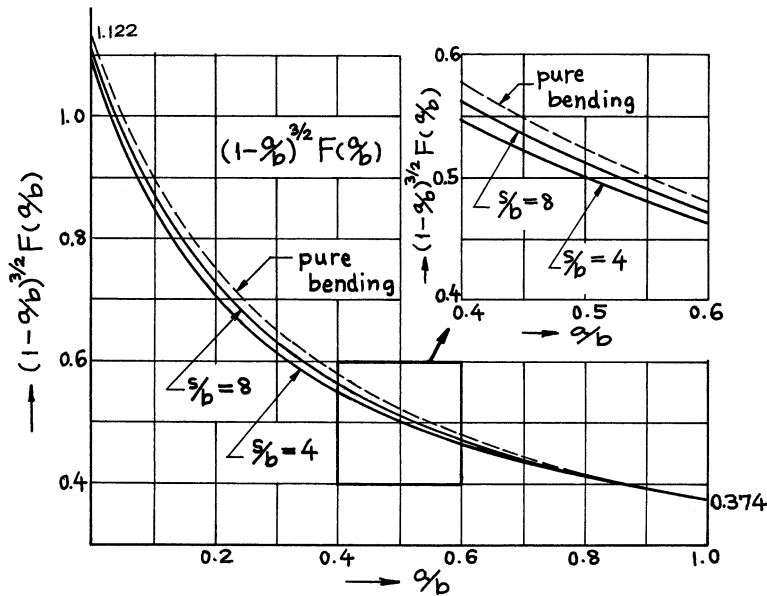
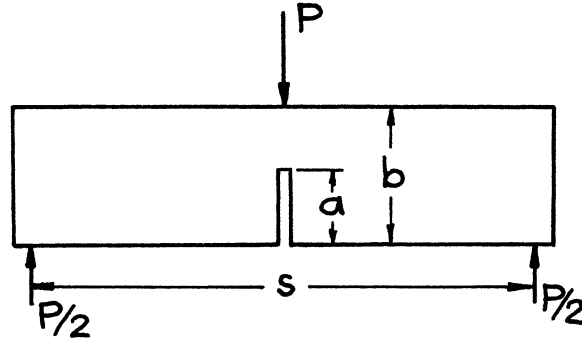
A. Stress Intensity Factor

$$\sigma = \frac{6M}{b^2} \left(M = \frac{Ps}{4} \right)$$

$$K_I = \sigma \sqrt{\pi a} F(a/b)$$

Numerical Values of $F(a/b)$

The curves in the following figure have 1% accuracy.



Methods and References

1. Boundary Collocation Method ($s/b = 4, 8$) (Gross 1965b)
2. Green's Function Method ($s/b = 3, 8$) (Emery 1969)

Empirical Formulas

- a. Accuracy
- b. Method, reference

For $s/b = 4$,

$$F(a/b) = \frac{1}{\sqrt{\pi}} \cdot \frac{1.99 - a/b(1 - a/b) \left(2.15 - 3.93 a/b + 2.7 (a/b)^2 \right)}{(1 + 2 a/b)(1 - a/b)^{3/2}}$$

- a. 0.5% for any a/b
- b. Srawley 1976

For $s/b = 8$,

$$F(a/b) = 1.106 - 1.552(a/b) + 7.71(a/b)^2 - 13.53(a/b)^3 + 14.23(a/b)^4$$

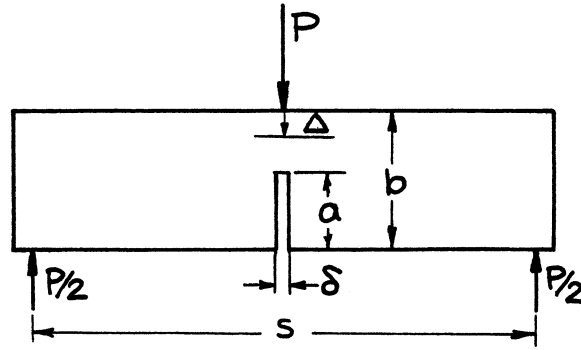
- a. 0.2% for $a/b \leq 0.6$
- b. Least squares fitting, **Brown 1966**

B. Displacements

Crack Opening at Edge

$$\delta = \frac{4\sigma a}{E'} V_1(a/b)$$

Gross' results (**Gross 1967**, Boundary Collocation Method, $s/b = 4$) are expected to have 0.5% accuracy for $0.2 \leq a/b \leq 0.7$. An empirical formula with 1% accuracy for any a/b for $s/b = 4$ is (**Tada 1973**)



$$V_1(a/b) = 0.76 - 2.28(a/b) + 3.87(a/b)^2 - 2.04(a/b)^3 + \frac{0.66}{(1 - a/b)^2}$$

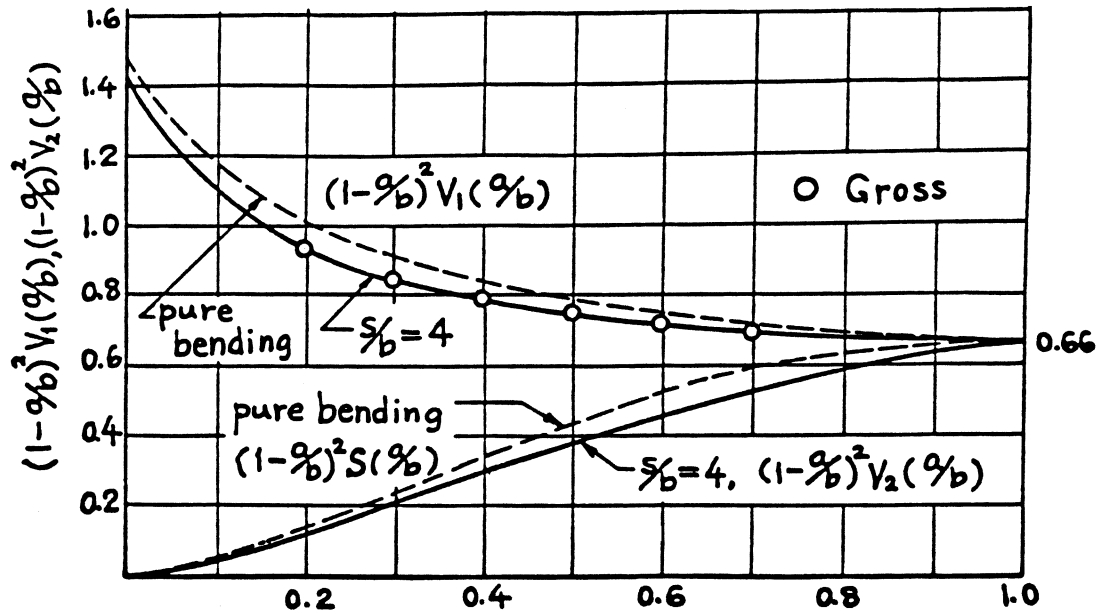
Additional Load Point Displacement due to Crack

$$\Delta_{crack} = \Delta_{total} - \Delta_{no\ crack}$$

$$\Delta_{crack} = \frac{\sigma}{E'} s V_2(a/b)$$

The following formula has better than 1% accuracy for any a/b ; for $s/b = 4$:

$$V_2(a/b) = \left(\frac{a/b}{1 - a/b} \right)^2 \left\{ 5.58 - 19.57(a/b) + 36.82(a/b)^2 - 34.94(a/b)^3 + 12.77(a/b)^4 \right\}$$



Method: Paris' Equation (Paris 1957) (see Appendix B)

Reference: Tada 1973

(See also pages 2.13, 2.27, 9.1 etc., for related solutions.)

Note: The curves for $s/b = 8$ are nearly averages between the curves for pure bending and $s/b = 4$. For other cases ($s/b > 4$), displacements can be estimated by interpolation with fair accuracy.

THE COMPACT TENSION TEST SPECIMEN

$$K_I = \sigma \sqrt{a} F_I(a/b, h/b, d/h)$$

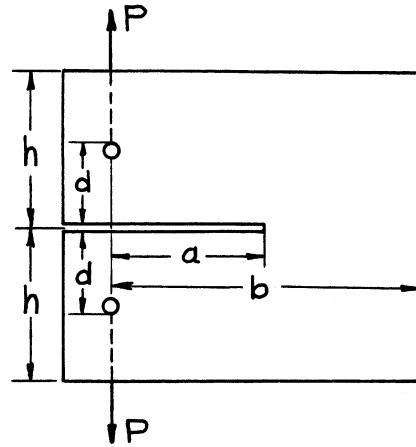
where $\sigma = P/b$

or $K_I = \sigma_N \sqrt{b-a} F_2(a/b, h/b, d/h)$

where $\sigma_N = \sigma_{N_{Tension}} + \sigma_{N_{Bending}}$

$$= \frac{P}{b-a} + \frac{6P \left(a + \frac{b-a}{2} \right)}{(b-a)^2}$$

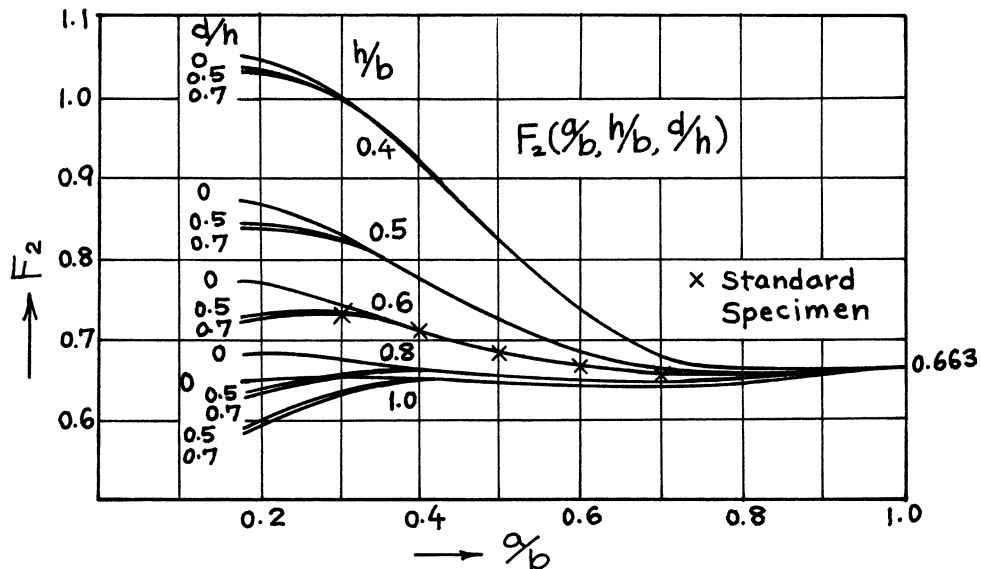
$$= \frac{2P(2b+a)}{(b-a)^2}$$



Numerical Values of F_2

$$\left(F_1 = \frac{2(2 + a/b)}{(1 - a/b)^{3/2}} \cdot \frac{1}{\sqrt{a/b}} \cdot F_2 \right)$$

The curves in the following figure have better than 1% accuracy.

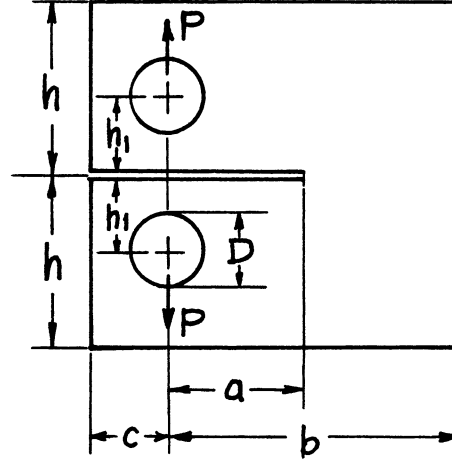


Method: Boundary Collocation Method
References: Gross 1970; Srawley 1972

Standard Specimen (ASTM Standard E-399-72)

Standard geometry of compact tension specimen is shown below.

$$\begin{aligned} h &= 0.6b \\ h_1 &= 0.275b \\ D &= 0.25b \\ c &= 0.25b \\ (\text{Thickness} &= b/2) \end{aligned}$$



A. Stress Intensity Factor

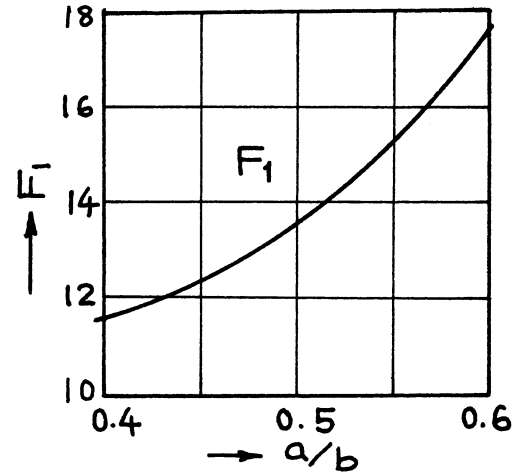
The F_2 values for the standard specimen are plotted in the previous graph. For the range $0.4 \leq a/b \leq 0.6$, the values of F_1 are plotted.

Note: $F_2 \simeq F_2(a/b, 0.6, 0.7)$

Empirical Formula

For the standard specimen, the following formula has 0.5% accuracy for $a/b > 0.2$ (Srawley 1976).

$$\begin{aligned} F_2(a/b) &= 0.443 + 2.32(a/b) \\ &\quad - 6.66(a/b)^2 + 7.36(a/b)^3 - 2.8(a/b)^4 \end{aligned}$$



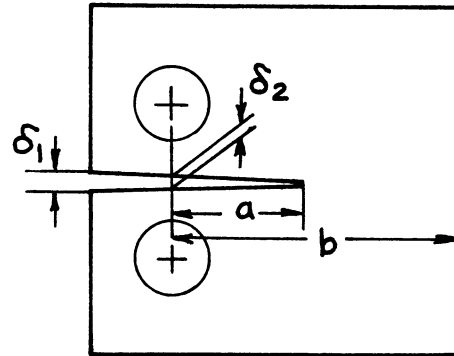
B. Displacements

Opening at Crack Edge

$$\delta_1 = \frac{P}{E'} V_1(a/b)$$

Opening at Loadline

$$\delta_2 = \frac{P}{E'} V_2(a/b)$$



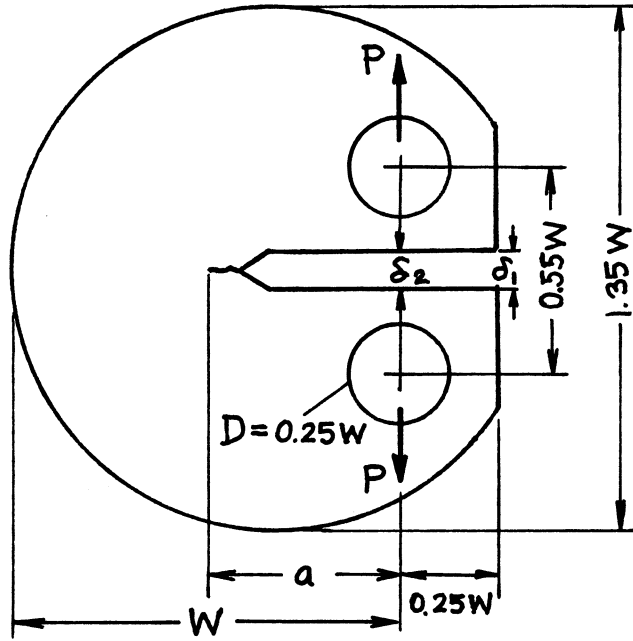
The following formulas for V_1 and V_2 for the standard specimen have 0.5% accuracies for $0.2 \leq a/b \leq 0.95$ (Saxena 1978).

$$V_1(a/b) = \left(1 + \frac{0.25}{a/b}\right) \left(\frac{1 + a/b}{1 - a/b}\right)^2 \left[1.6137 + 12.678(a/b) - 14.231(a/b)^2 - 16.610(a/b)^3 + 35.050(a/b)^4 - 14.494(a/b)^5\right]$$

$$V_2(a/b) = \left(\frac{1 + a/b}{1 - a/b}\right)^2 \left[2.1630 + 12.219(a/b) - 20.065(a/b)^2 - 0.9925(a/b)^3 + 20.609(a/b)^4 - 9.9314(a/b)^5\right]$$

Method: Boundary Collocation Method

THE ROUND (DISK-SHAPED) COMPACT SPECIMEN



thickness = B

$$A = \frac{a}{W}$$

$$\bar{\sigma} = \frac{P}{WB}$$

A. Stress Intensity Factor

$$K_I = \bar{\sigma} \sqrt{W} F(A)$$

$$F(A) = \frac{(2 + A) \left(0.76 + 4.8A - 11.58A^2 + 11.43A^3 - 4.08A^4 \right)}{(1 - A)^{3/2}}$$

B. Displacements

Opening Displacement at Edge

$$\delta_1 = \frac{\bar{\sigma} W}{E'} \cdot V_1(A)$$

$$V_1(A) = \exp \left(1.742 - 0.495A + 14.71A^2 - 22.06A^3 + 14.44A^4 \right)$$

Opening Displacement at Load Line

$$\delta_2 = \frac{\bar{\sigma} W}{E'} \cdot V_2(A)$$

$$V_2(A) = \exp \left(0.26 + 5.381A + 2.105A^2 - 8.853A^3 + 9.122A^4 \right)$$

Method: Boundary Collocation Method

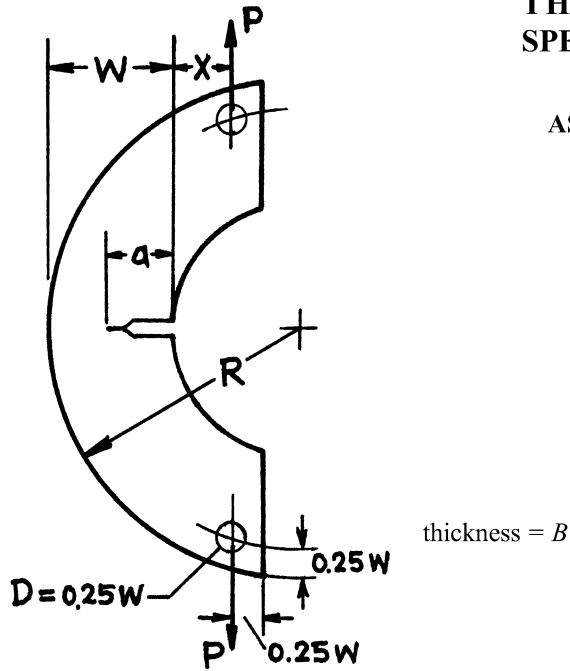
Accuracy: K_I 0.3% for $0.2 \leq A \leq 1.0$; δ_1 and δ_2 0.5% for $0.2 \leq A \leq 0.8$

References: Newman 1979a, 1981b

THE ARC-SHAPED (C-SHAPED) SPECIMEN

ASTM E-399 Standard Specimens:

$$X/W = 0 \text{ and } 0.5$$



$$A = \frac{a}{W} \quad \bar{\sigma} = \frac{P}{WB}$$

$$K_I = \bar{\sigma} \sqrt{\pi a} \cdot f\left(\frac{W}{R}, \frac{X}{W}, A\right) \cdot F(A)$$

$$f\left(\frac{W}{R}, \frac{X}{W}, A\right) = \left(3 \frac{X}{W} + 1.9 + 1.1A\right) \left[1 + 0.25(1-A)^2 \frac{W}{R}\right]$$

$$F(A) = \frac{3.74 - 6.30A + 6.32A^2 - 2.43A^3}{\sqrt{\pi}(1-A)^{3/2}}$$

Method: Boundary Collocation Method

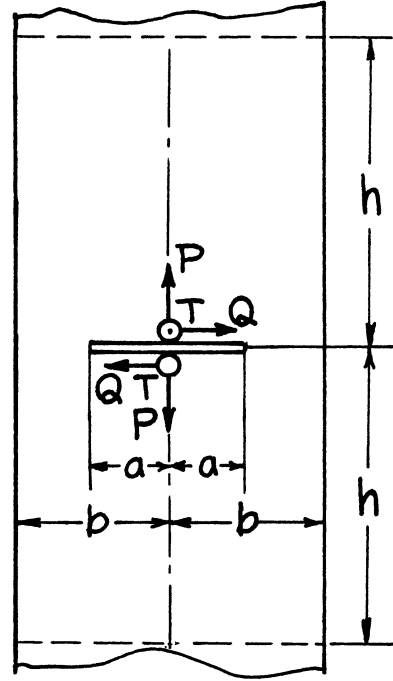
Accuracy: $X/W = 0$ and 0.5 : 1% for $0.45 \leq A \leq 0.55$; 1.5% for $0.2 \leq A \leq 1.0$.
 $0 \leq X/W \leq 1.0$: 3% for $0.2 \leq A \leq 1.0$

Reference: **Kapp 1980**

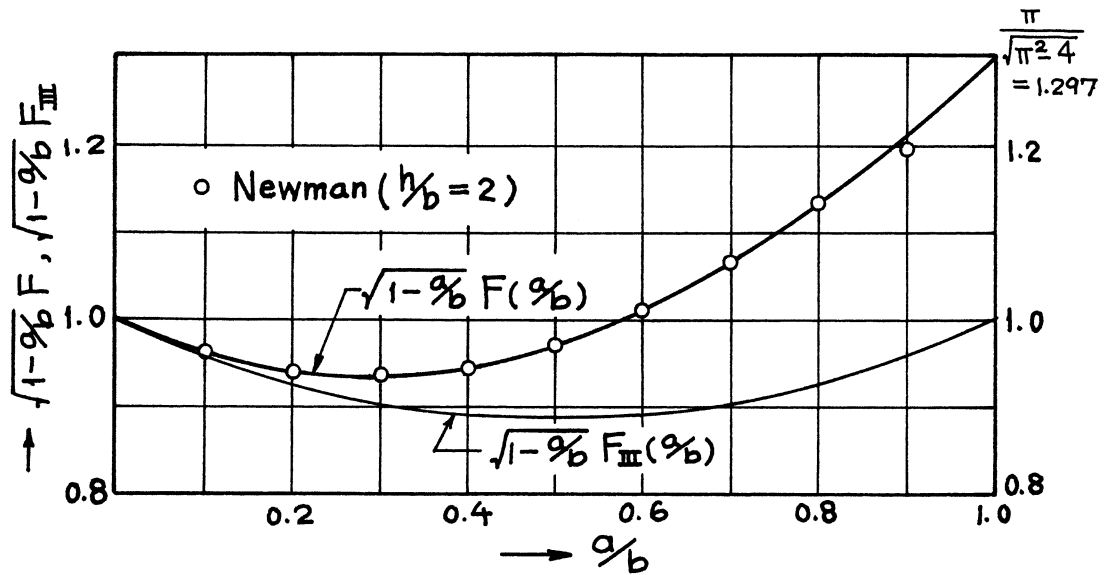
A. Stress Intensity Factor

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F(a/b) \\ F(a/b) \\ F_{III}(a/b) \end{Bmatrix}$$

$$F_{III}(a/b) = \sqrt{\frac{\pi a}{b}} \sin \frac{\pi a}{b}$$

Numerical Values of $F(a/b)$

Newman's results based on a method of boundary collocation are expected to have the accuracy of the order of 0.1% for $0.1 \leq a/b \leq 0.8$ (Newman 1971, $h/b = 2$).



Empirical Formulas

- Accuracy
- Method of derivation, reference

$$F(a/b) = \left(1.297 - 0.297 \cos \frac{\pi a}{2b}\right) \sqrt{\frac{\pi a}{b} \sin \frac{\pi a}{b}}$$

- Better than 1% for any a/b
- Asymptotic Interpolation, **Tada 1973**

$$F(a/b) = \frac{1 - 0.5(a/b) + 0.957(a/b)^2 - 0.16(a/b)^3}{\sqrt{1 - a/b}}$$

- Better than 0.3% for any a/b
- Modified asymptotic formula (**Tada 1973**)

B. Displacements at Remote Points ($h/b > 2$)

$$\begin{Bmatrix} \Delta_I \\ \Delta_{II} \end{Bmatrix} = \frac{2}{E'} \begin{Bmatrix} P \\ Q \end{Bmatrix} D(a/b)$$

$$\Delta_{III} = \frac{T}{G} \frac{2}{\pi} \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

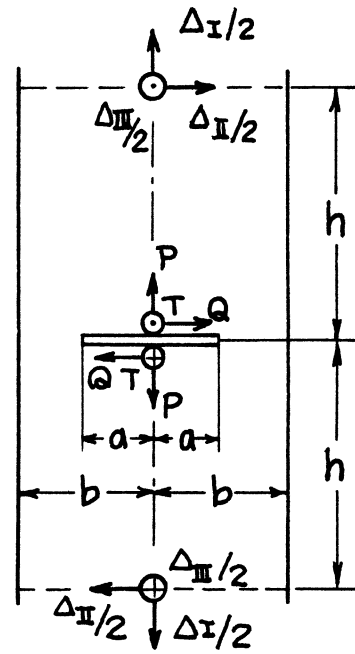
The following formula for $D(a/b)$ has better than 0.6% accuracy for any a/b .

$$D(a/b) = -0.071(a/b) - 0.535(a/b)^2 + 0.169(a/b)^3 - 0.090(a/b)^4 + 0.020(a/b)^5 - 1.071 \ln(1 - a/b)$$

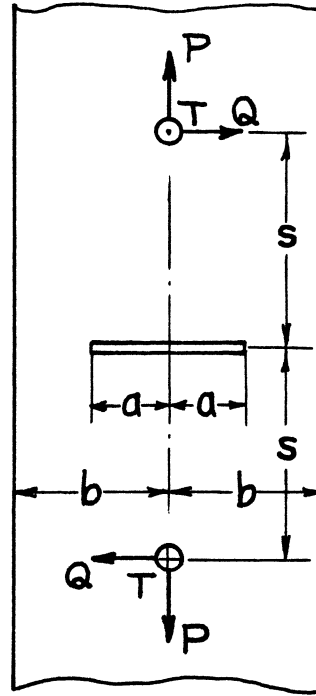
$$\text{where} \quad \left(1.071 = \frac{2\pi}{\pi^2 - 4} \right)$$

Method: Paris' Equation (**Paris 1957**) (see **Appendix B**)

Reference: **Tada 1973**



$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{1}{\sqrt{2b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F_I(a/b, s/b) \\ F_{II}(a/b, s/b) \\ F_{III}(a/b, s/b) \end{Bmatrix}$$



$$F_{III}(a/b, s/b) = \sqrt{\tan \frac{\pi a}{2b}} \cdot \frac{1}{\sqrt{1 - \left(\frac{\cos \frac{\pi a}{2b}}{\cosh \frac{\pi s}{2b}} \right)^2}}$$

$$\begin{Bmatrix} F_I(a/b, s/b) \\ F_{II}(a/b, s/b) \end{Bmatrix} = f(a/b, s/b) \begin{bmatrix} 1 \{ \pm \} \alpha \frac{\frac{\pi s}{2b} \tanh \frac{\pi s}{2b}}{\left(\cosh \frac{\pi s}{2b} \right)^2} \\ \left(\frac{\cosh \frac{\pi s}{2b}}{\cosh \frac{\pi a}{2b}} \right) - 1 \end{bmatrix} F_{III}(a/b, s/b)$$

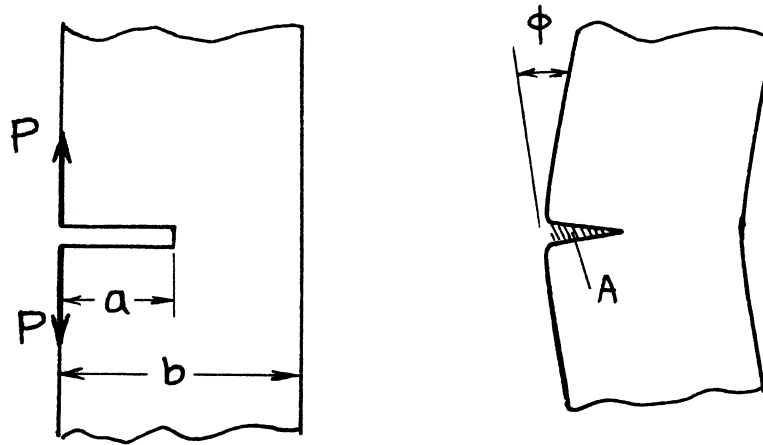
$$f(a/b, s/b) = 1 + \left\{ 0.297 - 0.115 \left(1 - \operatorname{sech} \frac{\pi s}{2b} \right) \sin \frac{\pi a}{b} \right\} \left(1 - \cos \frac{\pi a}{2b} \right)$$

$$\alpha = \begin{cases} \frac{1+\nu}{2} & \text{plane stress} \\ \frac{1}{2(1-\nu)} & \text{plane strain} \end{cases}$$

Method: Asymptotic Interpolation

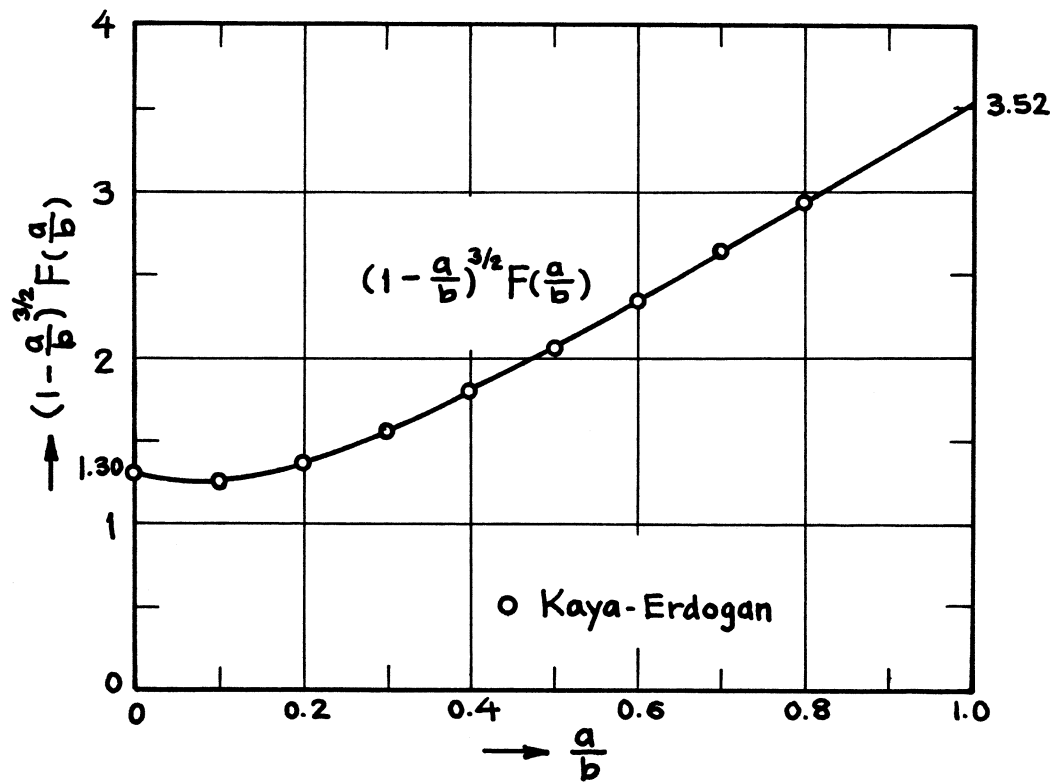
Accuracy: F_{III} exact; F_I, F_{II} better than 1% for any a/b and s/b

Reference: **Tada 1973**



$$K_I = \frac{2P}{\sqrt{\pi a}} F\left(\frac{a}{b}\right)$$

$$F\left(\frac{a}{b}\right) = \frac{.46 + 3.06 \frac{a}{b} + .84 \left(1 - \frac{a}{b}\right)^5 + .66 \left(\frac{a}{b}\right)^2 \left(1 - \frac{a}{b}\right)^2}{\left(1 - \frac{a}{b}\right)^{3/2}}$$



Crack Opening Area

$$A = \frac{4Pa}{E'} \cdot S\left(\frac{a}{b}\right)$$

$$S\left(\frac{a}{b}\right) = \frac{1.46 + 3.42\left(1 - \cos\frac{\pi a}{2b}\right)}{\left(\cos\frac{\pi a}{2b}\right)^2}$$

Rotation due to Crack

$$\phi = \frac{24Pa}{E' b^2} \cdot \Phi\left(\frac{a}{b}\right)$$

$$\Phi\left(\frac{a}{b}\right) = \frac{.878 - .218\frac{a}{b} + .58\left(1 - \frac{a}{b}\right)^5}{\left(1 - \frac{a}{b}\right)^2}$$

Methods: K Singular Integral Equation; A and ϕ Paris' Equation (see **Appendix B**)

Accuracy: K better than 0.5%; Empirical formula 1%; A and ϕ 1%

References: **Kaya 1980, Tada 1985**

$$K_I = \frac{2P}{\sqrt{\pi a}} \cdot \frac{G\left(\frac{c}{a}, \frac{a}{b}\right)}{\left(1 - \frac{a}{b}\right)^{3/2} \sqrt{1 - \left(\frac{c}{a}\right)^2}}$$

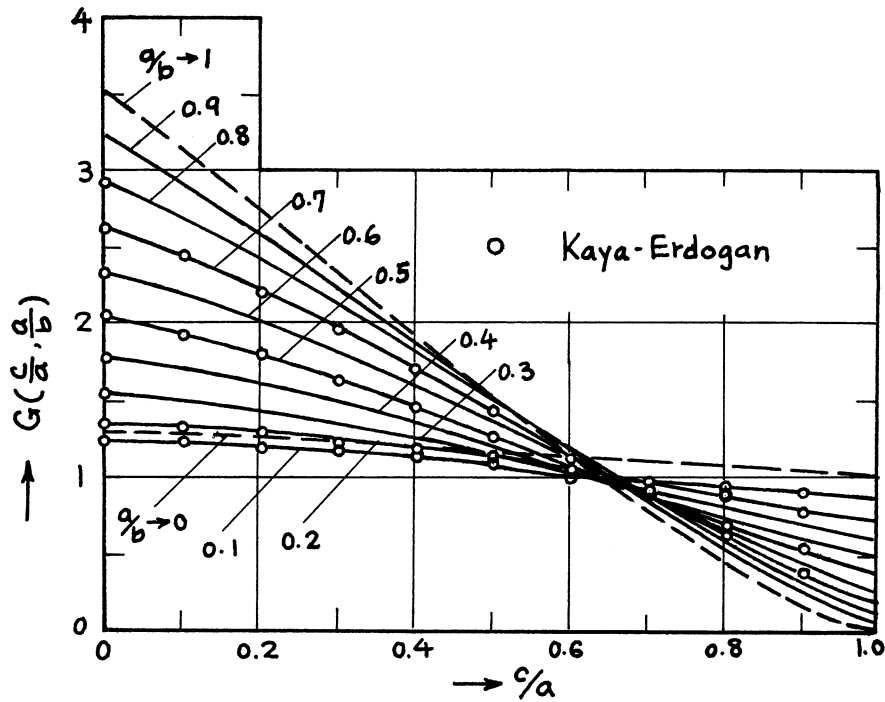
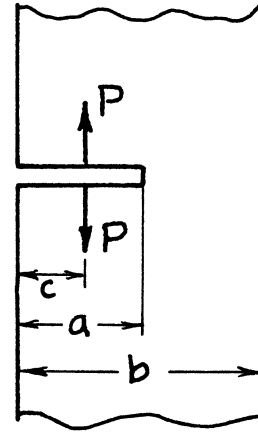
$$G\left(\frac{c}{a}, \frac{a}{b}\right) = g_1\left(\frac{a}{b}\right) + g_2\left(\frac{a}{b}\right) \cdot \frac{c}{a} + g_3\left(\frac{a}{b}\right) \cdot \left(\frac{c}{a}\right)^2 + g_4\left(\frac{a}{b}\right) \cdot \left(\frac{c}{a}\right)^3$$

$$g_1\left(\frac{a}{b}\right) = .46 + 3.06 \frac{a}{b} + .84\left(1 - \frac{a}{b}\right)^5 + .66\left(\frac{a}{b}\right)^2 \left(1 - \frac{a}{b}\right)^2$$

$$g_2\left(\frac{a}{b}\right) = -3.52\left(\frac{a}{b}\right)^2$$

$$g_3\left(\frac{a}{b}\right) = 6.17 - 28.22 \frac{a}{b} + 34.54\left(\frac{a}{b}\right)^2 - 14.39\left(\frac{a}{b}\right)^3 - \left(1 - \frac{a}{b}\right)^{3/2} - 5.88\left(1 - \frac{a}{b}\right)^5 - 2.64\left(\frac{a}{b}\right)^2 \left(1 - \frac{a}{b}\right)^2$$

$$g_4\left(\frac{a}{b}\right) = -6.63 + 25.16 \frac{a}{b} - 31.04\left(\frac{a}{b}\right)^2 + 14.41\left(\frac{a}{b}\right)^3 + 2\left(1 - \frac{a}{b}\right)^{3/2} + 5.04\left(1 - \frac{a}{b}\right)^5 + 1.98\left(\frac{a}{b}\right)^2 \left(1 - \frac{a}{b}\right)^2$$



Methods: Singular Integral Equation (Kaya-Erdogan, $a/b = .1, .2, .5, .7$). Estimated by Interpolation for other a/b .

Accuracy: 1% (Curves are based on the empirical formula above)

References: **Kaya 1980, Tada 1985**

NOTE: Dashed lines are $G(a/b \rightarrow 0) = 1.3 - .3(c/a)^{5/4}$ and $G(a/b \rightarrow 1) = 3.52(1 - c/a)\sqrt{1 - (c/a)^2}$

A. Stress Intensity Factor

$$\begin{Bmatrix} K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{\pi a}} \begin{Bmatrix} Q \\ T \end{Bmatrix} \begin{Bmatrix} F_{II}(a/b) \\ F_{III}(a/b) \end{Bmatrix}$$

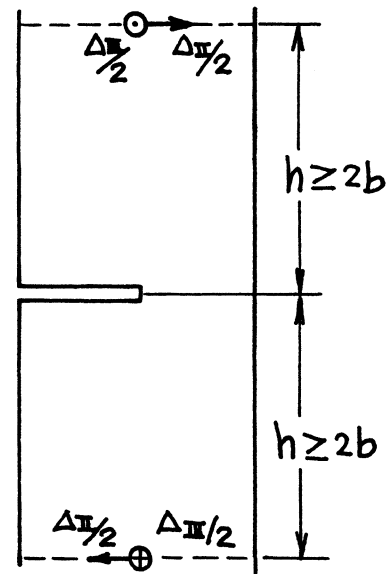
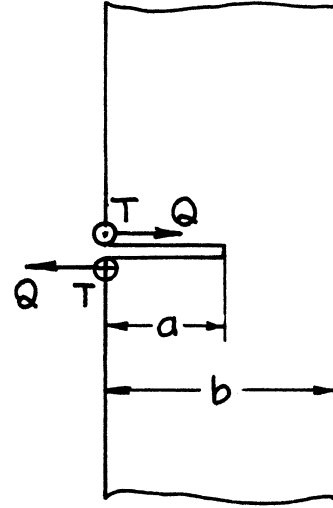
$$F_{II}(a/b) = \frac{1.30 - 0.65(a/b) + 0.37(a/b)^2 + 0.28(a/b)^3}{\sqrt{1 - a/b}}$$

$$F_{III}(a/b) = \sqrt{\frac{\pi a}{b} \sin \frac{\pi a}{b}}$$

Method: Asymptotic Interpolation

Accuracy: F_{II} better than 1% for any a/b ; F_{III} exact

Reference: **Tada 1973**



B. Displacements

$$\Delta_{II} = \frac{4Q}{E'} U(a/b)$$

$$\Delta_{III} = \frac{2T}{G} W(a/b)$$

$$U(a/b) = -0.184(a/b) - 0.637(a/b)^2 - 0.129(a/b)^3 + 0.026(a/b)^4 + 0.028(a/b)^5 + 0.008(a/b)^6 - 1.644 \ln(1 - a/b)$$

$$W(a/b) = \frac{2}{\pi} \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

Method: Paris' Equation (**Paris 1957**) (see **Appendix B**)

Accuracy: U better than 2% for any a/b ; W exact

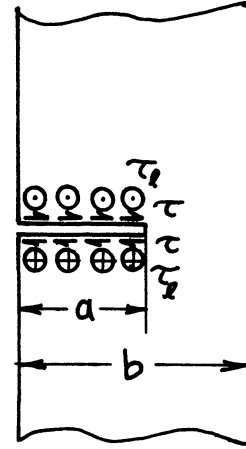
Reference: **Tada 1973**

A. Stress Intensity Factor

$$\begin{Bmatrix} K_{II} \\ K_{III} \end{Bmatrix} = \begin{Bmatrix} \tau \\ \tau_\ell \end{Bmatrix} \sqrt{\pi a} \begin{Bmatrix} F_{II}(a/b) \\ F_{III}(a/b) \end{Bmatrix}$$

$$F_{II}(a/b) = \frac{1.122 - 0.561(a/b) + 0.085(a/b)^2 + 0.180(a/b)^3}{\sqrt{1 - a/b}}$$

$$F_{III}(a/b) = \sqrt{\frac{2b}{\pi a}} \tan \frac{\pi a}{2b}$$



Method: Asymptotic Interpolation

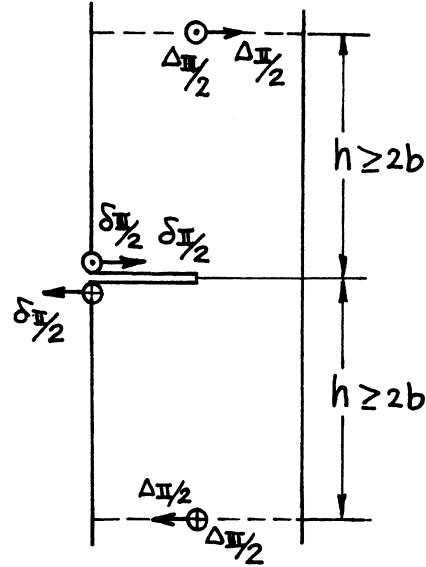
Accuracy: F_{II} better than 2% for any a/b ; F_{III} exact

Reference: **Tada 1973**

B. Displacements

$$\begin{Bmatrix} \delta_{II} \\ \Delta_{II} \end{Bmatrix} = \frac{4\tau a}{E'} \begin{Bmatrix} U_1(a/b) \\ U_2(a/b) \end{Bmatrix}$$

$$\begin{Bmatrix} \delta_{III} \\ \Delta_{III} \end{Bmatrix} = \frac{2\tau_\ell a}{G} \begin{Bmatrix} W_1(a/b) \\ W_2(a/b) \end{Bmatrix}$$



$$U_1(a/b) = -0.184 - 0.637(a/b) - 0.129(a/b)^2 + 0.026(a/b)^3 + 0.028(a/b)^4 + 0.008(a/b)^5 - 1.644(b/a) \ln(1 - a/b)$$

$$U_2(a/b) = 1.46(a/b) - 0.259(a/b)^2 - 0.091(a/b)^3 + 0.052(a/b)^4 - 0.019(a/b)^5 - 0.008(a/b)^6 - 0.518 \ln(1 - a/b)$$

$$W_1(a/b) = \frac{2}{\pi} \frac{1}{(a/b)} \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

$$W_2(a/b) = \frac{2}{\pi} \frac{1}{(a/b)} \ln \left(\sec \frac{\pi a}{2b} \right)$$

Method: Paris' Equation (**Paris 1957**) (see **Appendix B**)

Accuracy: U_1 , U_2 better than 2% for any a/b ; W_1 , W_2 exact

Reference: **Tada 1973**

A. Stress Intensity Factor

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F(a/b) \\ F(a/b) \\ F_{III}(a/b) \end{Bmatrix}$$

$$F(a/b) = \frac{1.30 - 0.65(a/b) - 0.10(a/b)^2 + 0.45(a/b)^3}{\sqrt{1 - a/b}}$$

$$\text{or} \quad F(a/b) = \left(1 + 0.30 \cos^2 \frac{\pi a}{2b}\right) \sqrt{\frac{\pi a}{b} \sin \frac{\pi a}{b}}$$

$$F_{III}(a/b) = \sqrt{\frac{\pi a}{b} \sin \frac{\pi a}{b}}$$

Method: Asymptotic Interpolation

Accuracy: K_I better than 2% for any a/b ; F_{III} exact

Reference: **Tada 1973**

B. Displacements at Remote Points ($h/b > 2$)

$$\begin{Bmatrix} \Delta_I \\ \Delta_{II} \end{Bmatrix} = \frac{4}{E'} \begin{Bmatrix} P \\ Q \end{Bmatrix} V(a/b)$$

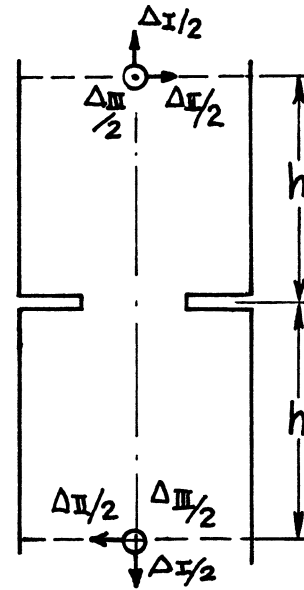
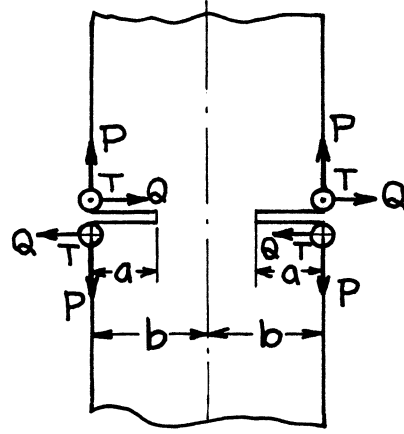
$$\Delta_{III} = \frac{2}{G} T \cdot \frac{2}{\pi} \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

The following formula for $V(a/b)$ has better than 2% for any a/b .

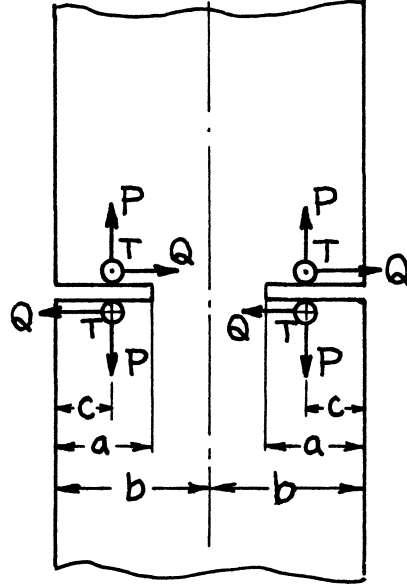
$$V(a/b) = 0.292 \left(\sin \frac{\pi a}{2b} \right) - 0.041 \left(\sin \frac{\pi a}{2b} \right)^3 - 0.004 \left(\sin \frac{\pi a}{2b} \right)^5 + 0.637 (a/b) \cosh^{-1} \left(\sec \frac{\pi a}{2b} \right)$$

Method: Paris' Equation (**Paris 1957**) (see **Appendix B**)

Reference: **Tada 1973**



$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{2b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F(a/b, c/a) \\ F(a/b, c/a) \\ F_{III}(a/b, c/a) \end{Bmatrix}$$



$$F_{III}(a/b, c/a) = \sqrt{\tan \frac{\pi a}{2b}} \frac{1}{\sqrt{1 - \left(\cos \frac{\pi a}{2b} / \cos \frac{\pi c}{2b} \right)^2}}$$

$$F(a/b, c/a) = \{1 + f(c/a) \cdot g(a/b)\} F_{III}(a/b, c/a)$$

where

$$f(c/a) = 0.3 \left\{ 1 - (c/a)^{5/4} \right\}$$

$$g(a/b) = 0.5 \left(1 - \sin \frac{\pi a}{2b} \right) \left(2 + \sin \frac{\pi a}{2b} \right)$$

Method: Asymptotic Interpolation

Accuracy: F_{III} exact; F better than 1% for any a/b and c/a

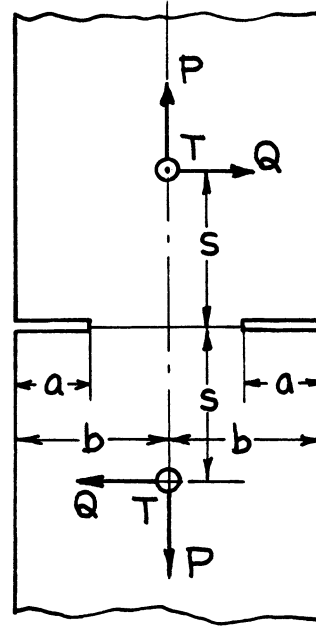
Reference: **Tada 1985**

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{1}{\sqrt{2b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F_I(a/b, s/b) \\ F_{II}(a/b, s/b) \\ F_{III}(a/b, s/b) \end{Bmatrix}$$

$$F_{III}(a/b, s/b) = \sqrt{\tan \frac{\pi a}{2b}} \frac{1}{\sqrt{1 + \left(\frac{\cos \frac{\pi a}{2b}}{\sin \frac{\pi s}{2b}} \right)^2}}$$

$$\begin{Bmatrix} F_I(a/b, s/b) \\ F_{II}(a/b, s/b) \end{Bmatrix} = \left(1 + 0.122 \cos^2 \frac{\pi a}{2b} \right) \begin{bmatrix} 1\{ \mp \} \alpha \frac{\frac{\pi s}{2b} \coth \frac{\pi s}{2b}}{\left(\frac{\sinh \frac{\pi s}{2b}}{\cos \frac{\pi a}{2b}} \right)^2} \\ 1 + \left(\frac{\sinh \frac{\pi s}{2b}}{\cos \frac{\pi a}{2b}} \right)^2 \end{bmatrix} F_{III}(a/b, s/b)$$

$$\alpha = \begin{cases} \frac{1+\nu}{2} & \text{plane stress} \\ \frac{1}{2(1-\nu)} & \text{plane strain} \end{cases}$$

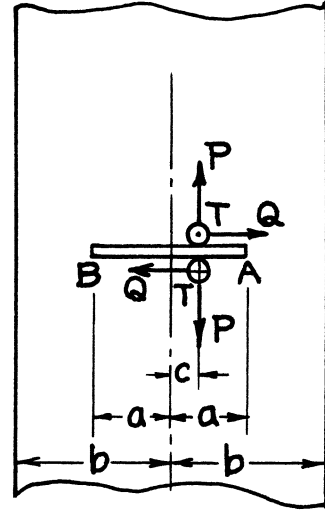


Method: Asymptotic Interpolation

Accuracy: F_I , F_{II} better than 2% for any a/b and s/b ; F_{III} exact

Reference: **Tada 1973**

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix}_{\frac{A}{B}} = \frac{1}{\sqrt{2b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F(a/b, c/a) \\ F(a/b, c/a) \\ F_{III}(a/b, c/a) \end{Bmatrix}_{\frac{A}{B}}$$



$$\begin{Bmatrix} F_{III_A}(a/b, c/a) \\ F_{III_B}(a/b, c/a) \end{Bmatrix} = \sqrt{\tan \frac{\pi a}{2b}} \cdot \frac{1 \begin{Bmatrix} + \\ - \end{Bmatrix} \begin{pmatrix} \sin \frac{\pi c}{2b} \\ \sin \frac{\pi a}{2b} \end{pmatrix}}{\sqrt{1 - \left(\frac{\cos \frac{\pi a}{2b}}{\cos \frac{\pi c}{2b}} \right)^2}}$$

$$\{F(a/b, c/a)\}_{\frac{A}{B}} = \left\{ 1 + 0.297 \sqrt{1 - (c/a)^2} \left(1 - \cos \frac{\pi a}{2b} \right) \right\} \{F_{III}(a/b, c/a)\}_{\frac{A}{B}}$$

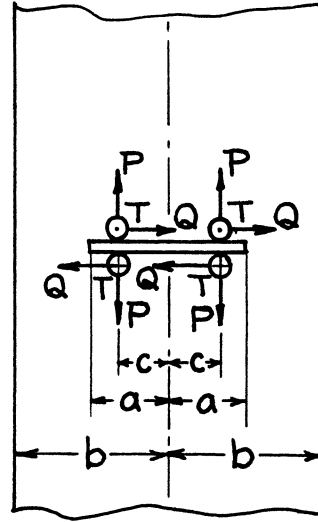
$$\text{where} \quad \left(0.297 = \frac{\pi}{\sqrt{\pi^2 - 4}} - 1 \right)$$

Method: Asymptotic Interpolation

Accuracy: F_{III} exact; F better than 1% for any a/b and c/a

Reference: **Tada 1973**

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{2b}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \begin{Bmatrix} F(a/b, c/a) \\ F(a/b, c/a) \\ F_{III}(a/b, c/a) \end{Bmatrix}$$



$$F_{III}(a/b, c/a) = \sqrt{\tan \frac{\pi a}{2b}} \cdot \frac{1}{\sqrt{1 - \left(\frac{\cos \frac{\pi a}{2b}}{\cos \frac{\pi c}{2b}} \right)^2}}$$

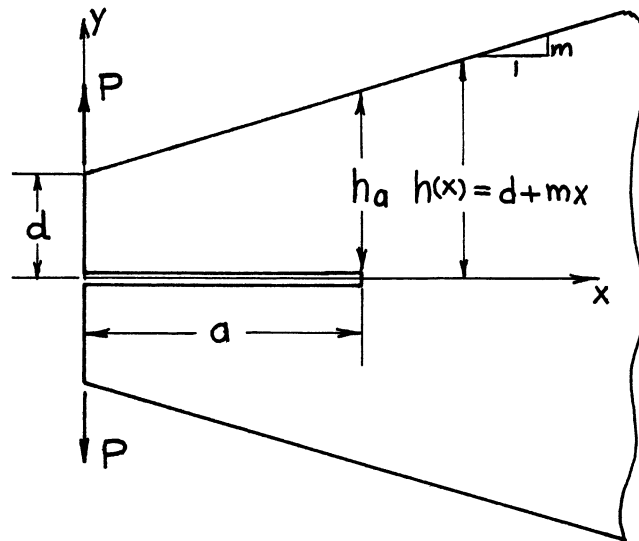
$$F(a/b, c/a) = \left\{ 1 + 0.297 \sqrt{1 - (c/a)^2} \left(1 - \cos \frac{\pi a}{2b} \right) \right\} F_{III}(a/b, c/a)$$

$$\text{where} \quad \left(0.297 = \frac{\pi}{\sqrt{\pi^2 - 4}} - 1 \right)$$

Method: Asymptotic Interpolation

Accuracy: F_{III} exact; F better than 1% for any a/b and c/a

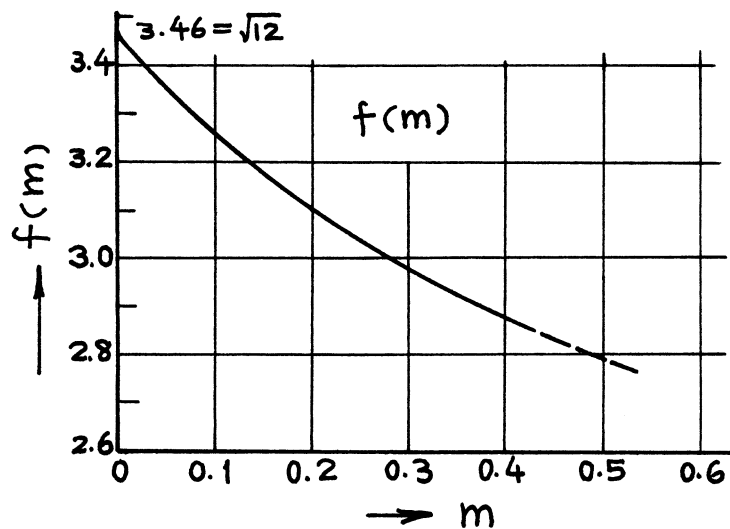
Reference: **Tada 1973**



$$h_a = h(a) = d + ma$$

$$K_I = \frac{P}{\sqrt{h_a}} \cdot f(m) \cdot \left(\frac{a}{h_a} + 0.7 \right)$$

$$f(m) = 3.46 - 2.65m + 1.89m^{3/2} \quad (m < 0.5)$$

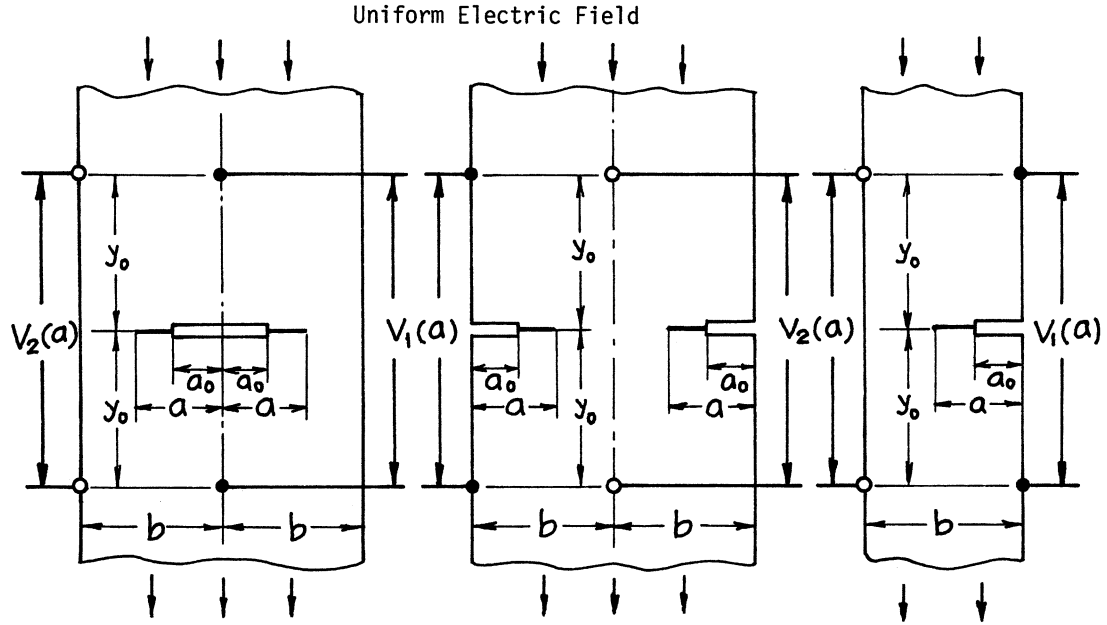


Method: Empirical formula based on the results by Boundary Collocation Method

Accuracy: Order of 1% for $a/h_a > 1$

References: Gross 1966; Srawley 1967; Tada 2000

ELECTRICAL POTENTIAL CALIBRATION



Electric Potentials: $V_1(a)$, $V_2(a)$

Potential Ratios:

$$\frac{V_1(a)}{V_1(a_0)} = \frac{\cosh^{-1} \left(\frac{\cosh \frac{\pi y_0}{2b}}{\cos \frac{\pi a}{2b}} \right)}{\cosh^{-1} \left(\frac{\cosh \frac{\pi y_0}{2b}}{\cos \frac{\pi a_0}{2b}} \right)}$$

$$\frac{V_2(a)}{V_2(a_0)} = \frac{\sinh^{-1} \left(\frac{\sinh \frac{\pi y_0}{2b}}{\cos \frac{\pi a}{2b}} \right)}{\sinh^{-1} \left(\frac{\sinh \frac{\pi y_0}{2b}}{\cos \frac{\pi a_0}{2b}} \right)}$$

Method: Conjugate Functions Method

Accuracy: Exact

References: **Johnson 1965; Tada 1973**